

# Service Manual

## TABLE OF CONTENTS

|                                     | PAGE |
|-------------------------------------|------|
| SAFETY CONSIDERATIONS.....          | 1    |
| INTRODUCTION.....                   | 1    |
| INDOOR UNIT SIZES .....             | 2    |
| MODEL NUMBER NOMENCLATURE.....      | 2    |
| ELECTRICAL DATA .....               | 3    |
| WIRING.....                         | 3    |
| CONNECTION DIAGRAMS.....            | 4    |
| WIRING DIAGRAMS.....                | 5    |
| REFRIGERANT CYCLE DIAGRAMS .....    | 6    |
| SYSTEM EVACUATION AND CHARGING..... | 8    |
| TROUBLESHOOTING.....                | 15   |
| DIAGNOSIS AND SOLUTION .....        | 17   |
| DISASSEMBLY INSTRUCTIONS .....      | 48   |
| APPENDICES .....                    | 55   |

## SAFETY CONSIDERATIONS

Installing, starting up, and servicing air-conditioning equipment can be hazardous due to unit pressures, electrical components, and equipment location (roofs, elevated structures, etc.).

Only trained, qualified installers and service mechanics should install, start-up, and service this equipment.

Untrained personnel can perform basic maintenance functions such as coil cleaning. All other operations should be performed by trained service personnel **only**.

When working on the equipment, observe the precautions in the literature and on tags, stickers, and labels attached to the equipment.

Follow all safety codes. Wear safety glasses and work gloves. Keep a quenching cloth and fire extinguisher nearby when brazing. Use care in handling, rigging, and setting bulky equipment.

Read this manual thoroughly and follow all the warnings or cautions included in the literature and attached to the unit. Consult local building codes and the National Electrical Code (NEC) for special requirements.

Recognize safety information. This is the safety-alert symbol . When you see this symbol on the unit and in instructions or manuals, be alert to the potential for personal injury. Understand these signal words: **DANGER**, **WARNING**, and **CAUTION**.

These words are used with the safety-alert symbol. **DANGER** identifies the most serious of hazards which will result in severe personal injury or death. **WARNING** signifies hazards which could result in personal injury or death. **CAUTION** is used to identify unsafe practices which may result in minor personal injury or product and property damage. **NOTE** is used to highlight suggestions which will result in enhanced installation, reliability, or operation.



## WARNING

### ELECTRICAL SHOCK HAZARD

Failure to follow this warning could result in personal injury or death. Before installing, modifying, or servicing the unit, the main electrical disconnect switch must be in the **OFF** position. There may be more than 1 disconnect switch. Lock out and tag the switch with a suitable warning label.



## WARNING



### EXPLOSION HAZARD

Failure to follow this warning could result in death, serious personal injury, and/or property damage.

Never use air or gases containing oxygen for leak testing or operating refrigerant compressors. Pressurized mixtures of air or gases containing oxygen can lead to an explosion.



## CAUTION

### EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation. **DO NOT** bury more than 36 in. (914 mm) of refrigerant pipe in the ground. If any section of pipe is buried, there must be a 6 in. (152 mm) vertical rise to the valve connections on the outdoor units. If more than the recommended length is buried, refrigerant may migrate to the cooler buried section during the extended periods of unit shutdown. This causes refrigerant slugging and could possibly damage the compressor at start-up.

## INTRODUCTION

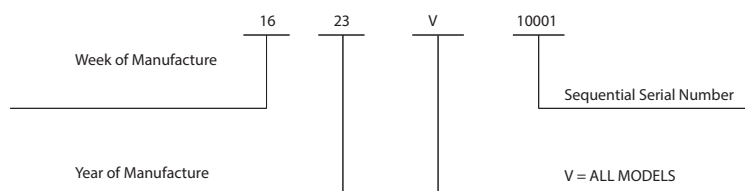
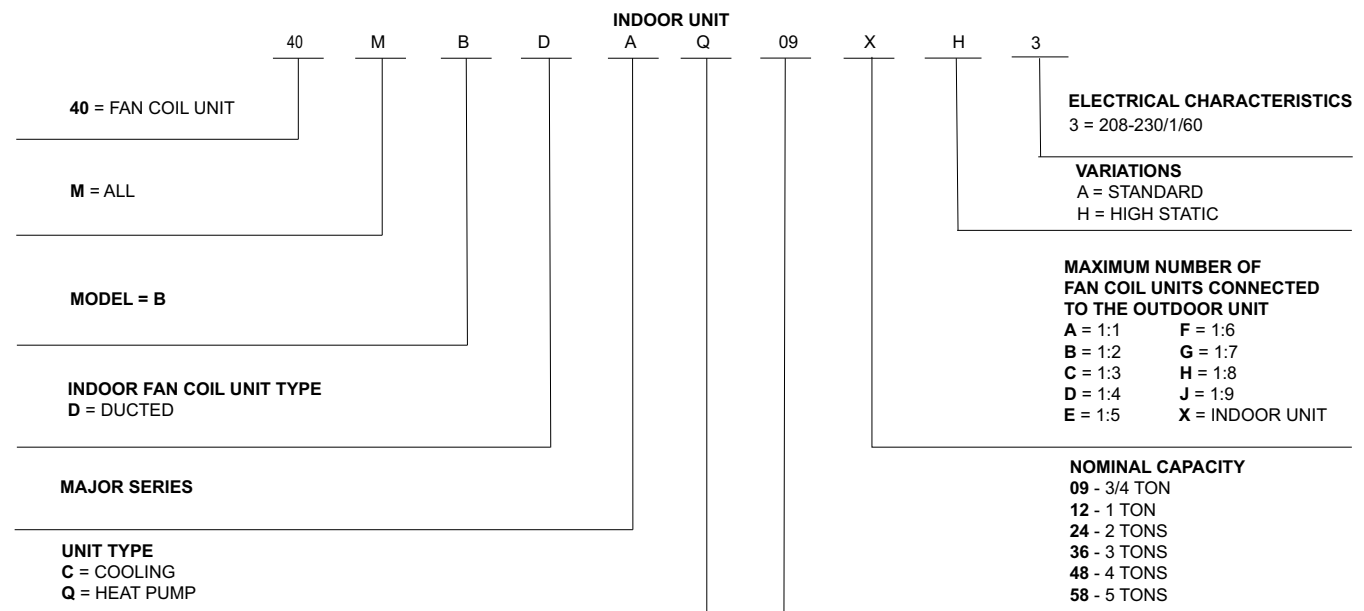
This service manual provides the necessary information to service, repair, and maintain the indoor units. This manual has an appendix (see “APPENDICES” on page 55 for data needed to perform troubleshooting tasks). Use the “TABLE OF CONTENTS” on page 1, to locate a desired topic.

## INDOOR UNIT SIZES

Table 1 — Indoor Unit Size

| KBTUH | V-PH-HZ    | MODEL NO.    |
|-------|------------|--------------|
| 9     | 208/230-60 | 40MBDAQ09XH3 |
| 12    |            | 40MBDAQ12XH3 |
| 18    |            | 40MBDAQ18XH3 |
| 24    |            | 40MBDAQ24XH3 |
| 36    |            | 40MBDAQ36XH3 |
| 48    |            | 40MBDAQ48XH3 |
| 58    |            | 40MBDAQ58XH3 |

## MODEL NUMBER NOMENCLATURE



Use of the AHRI Certified TM Mark indicates a manufacturer's participation in the program. For verification of certification for individual products, go to [www.ahridirectory.org](http://www.ahridirectory.org).



## ELECTRICAL DATA

**Table 2 — Electrical Data**

| INDOOR UNITS | V-PH-HZ      | INDOOR FAN |       |       |                         | MAX FUSE CB AMP                                                                                     |
|--------------|--------------|------------|-------|-------|-------------------------|-----------------------------------------------------------------------------------------------------|
|              |              | FLA        | HP    | W_Out | SYSTEM POWER FACTOR (%) |                                                                                                     |
| 9K           | 208-230/1/60 | 0.8        | 1/5   | 160   | 62.4                    | Refer to the outdoor unit installation instructions.<br>Indoor unit is powered by the outdoor unit. |
| 12K          |              | 1.1        | 1/5   | 160   | 97.2                    |                                                                                                     |
| 18K          |              | 1.8        | 1/5   | 160   | 71                      |                                                                                                     |
| 24K          |              | 2.2        | 1/5   | 160   | 89.6                    |                                                                                                     |
| 36K          |              | 2.2        | 3/4   | 560   | 98.6                    |                                                                                                     |
| 48K          |              | 2.9        | 3/4   | 560   | 99.1                    |                                                                                                     |
| 58K          |              | 2.5        | 15/16 | 700   | 99.3                    |                                                                                                     |

### LEGEND

FLA - Full Load Amps

## WIRING

All wires must be sized per NEC (National Electrical Code) or CEC (Canadian Electrical Code) and local codes. Use Electrical Data table MCA (minimum circuit amps) and MOCP (maximum over current protection) to correctly size the wires and the disconnect fuse or breakers respectively.

### SIZES 09-24 RECOMMENDED CONNECTION METHOD FOR POWER AND COMMUNICATION WIRING

**Power and Communication Wiring:** The main power is supplied to the outdoor unit. The field supplied 14/3 power/communication wiring, from the outdoor unit to the indoor unit, consists of four (4) wires and provides the power for the indoor unit. Two wires are high voltage AC power, one is communication wiring and the other is a ground wire.

**To minimize communication interference:** If installed in a high Electromagnetic field (EMF) area and communication issues arise, a 14/2 stranded shielded wire can be used to replace L2 and (S) between the outdoor and indoor units - landing the shield onto the ground in the outdoor unit only.

### WIRING SIZES 09-24

| CABLE            | CABLE SIZE | REMARKS                                                        |
|------------------|------------|----------------------------------------------------------------|
| Connection Cable | 14AWG      | 3 wire + Ground 1Φ 208/230 V<br>(Stranded wire is recommended) |

### SIZES 36-58 RECOMMENDED CONNECTION METHOD FOR POWER AND COMMUNICATION WIRING

**Power and Communication Wiring:** The main power is supplied to the outdoor unit. The field supplied power wiring from the outdoor unit to the indoor unit consists of three (3) wires and provides the power for the indoor unit. Two wires are high voltage AC power and one is a ground wire. To minimize voltage drop, the factory recommended wire size is 14/2 stranded with a ground.

**Communication Wiring:** A separate shielded stranded copper conductor only, with a 600 volt rating and double insulated copper wire, must be used as the communication wire from the outdoor unit to the indoor unit. Please use a separate shielded 16GA stranded control wire.

### WIRING SIZES 36-58

| CABLE                  | CABLE SIZE | REMARKS                               |
|------------------------|------------|---------------------------------------|
| Power Connection Cable | 14AWG      | 2 wire + Ground 1Φ 208/230V           |
| Communication Cable    | 16AWG      | 2 wire stranded shielded control wire |



## CAUTION

### EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation. Wires should be sized based on NEC and local codes.



## CAUTION

### EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.



## CAUTION

### EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

Be sure to comply with local codes while running wire from the indoor unit to the outdoor unit.

Every wire must be connected firmly. Loose wiring may cause the terminal to overheat or result in unit malfunction. A fire hazard may also exist. Ensure all wiring is tightly connected.

No wire should touch the refrigerant tubing, compressor or any moving parts.

Disconnecting means must be provided and shall be located within sight and readily accessible from the air conditioner.

CONNECTION DIAGRAMS

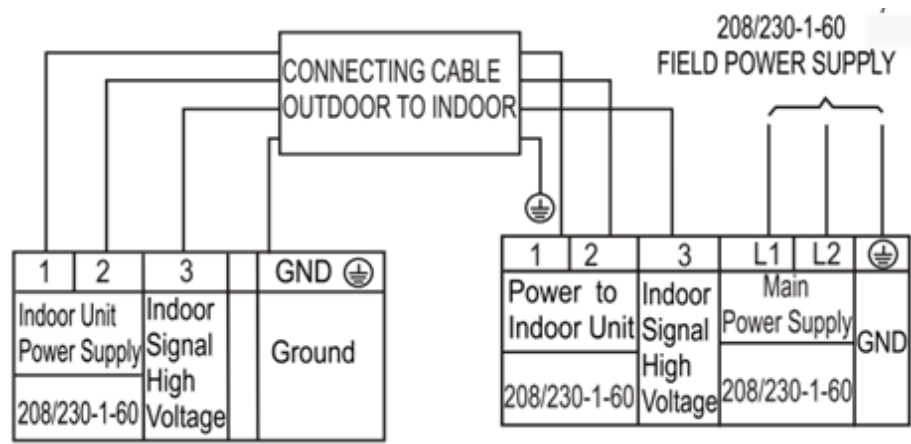


Fig. 1 — Connection Diagram - Sizes 9K, 12K, 18K, 24K

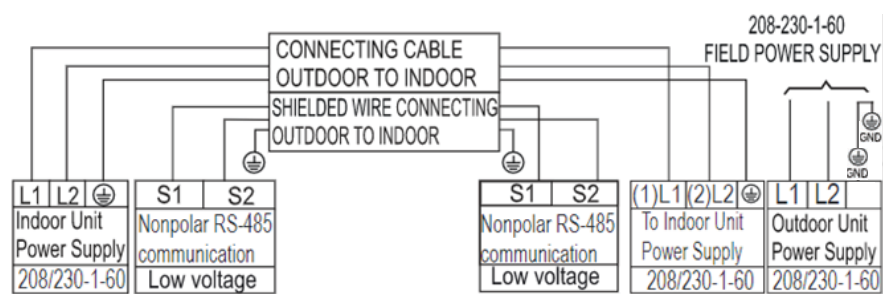


Fig. 2 — Connection Diagram - Size 36K

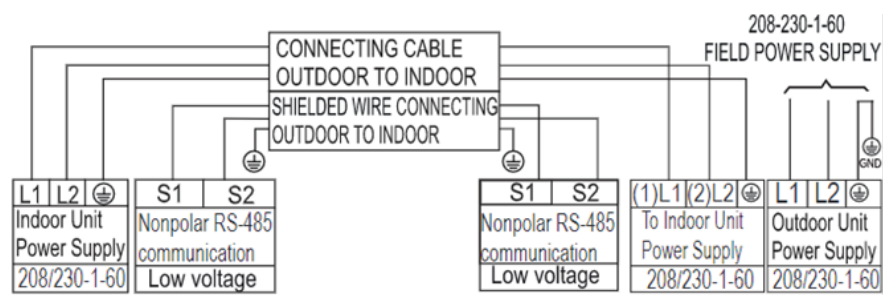


Fig. 3 — Connection Diagram - Sizes 48K to 58K

# WIRING DIAGRAMS

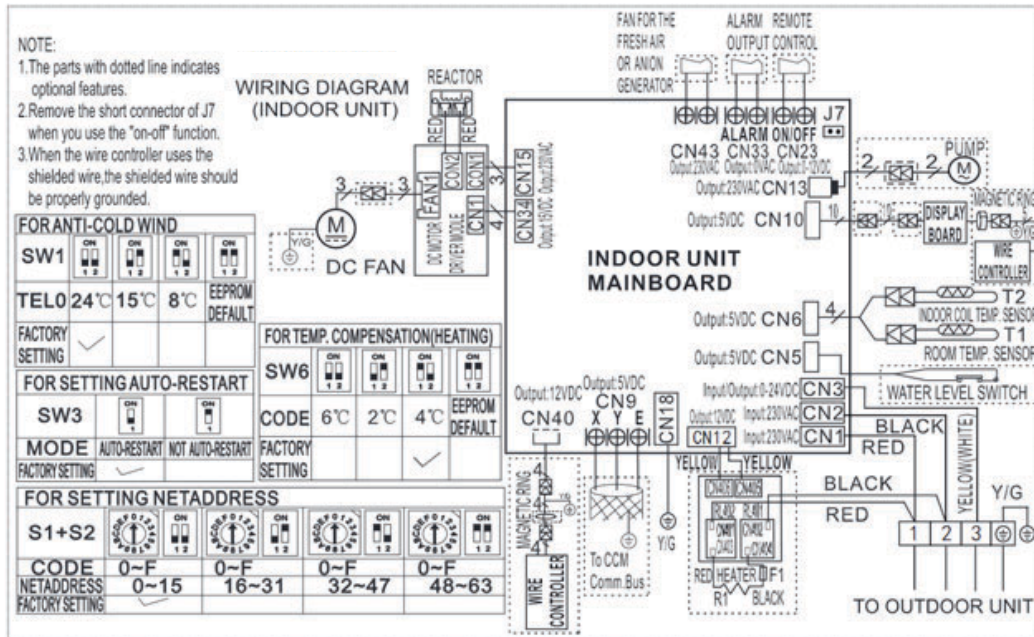


Fig. 4 — Wiring Diagram - Sizes 9K, 12K, 18K, 24K

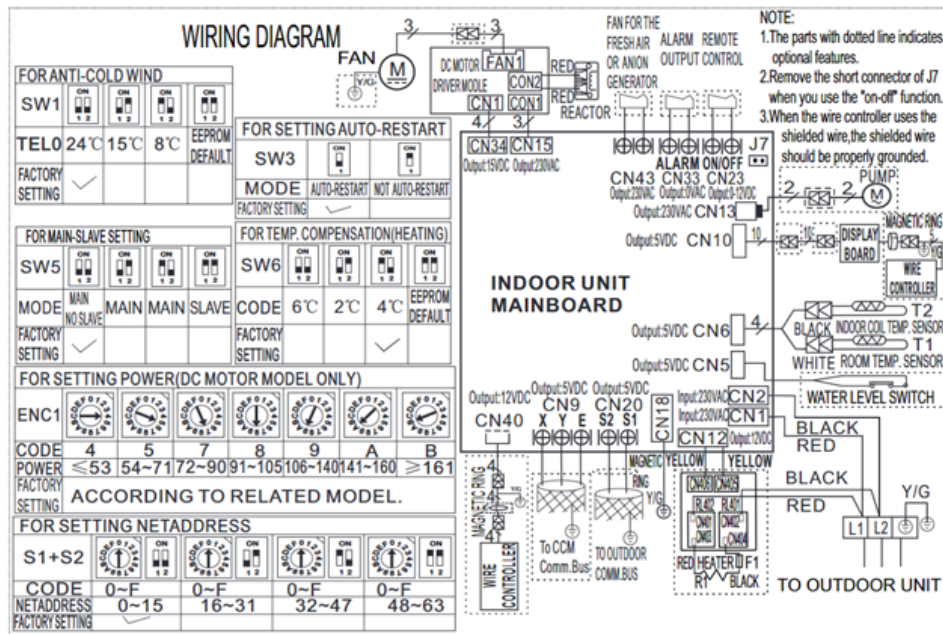
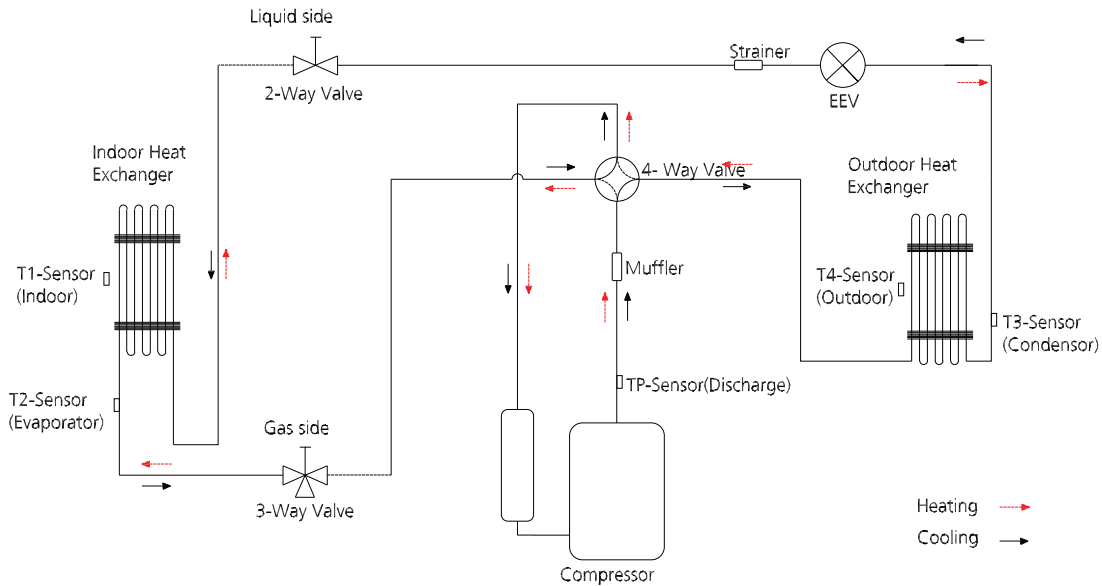
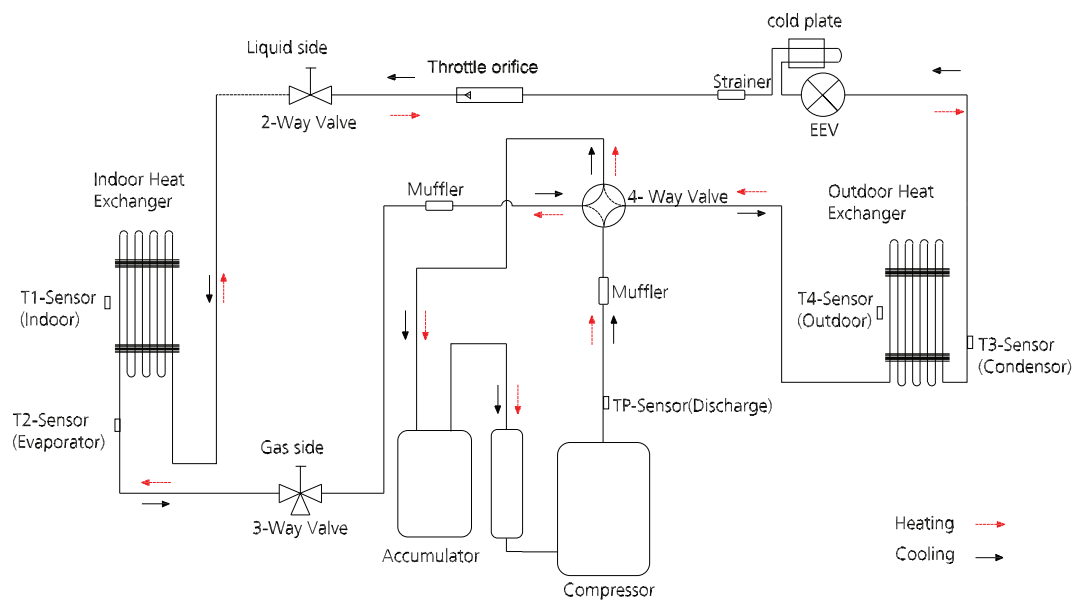


Fig. 5 — Wiring Diagram - Sizes 36K, 48K, 58K

## REFRIGERANT CYCLE DIAGRAMS

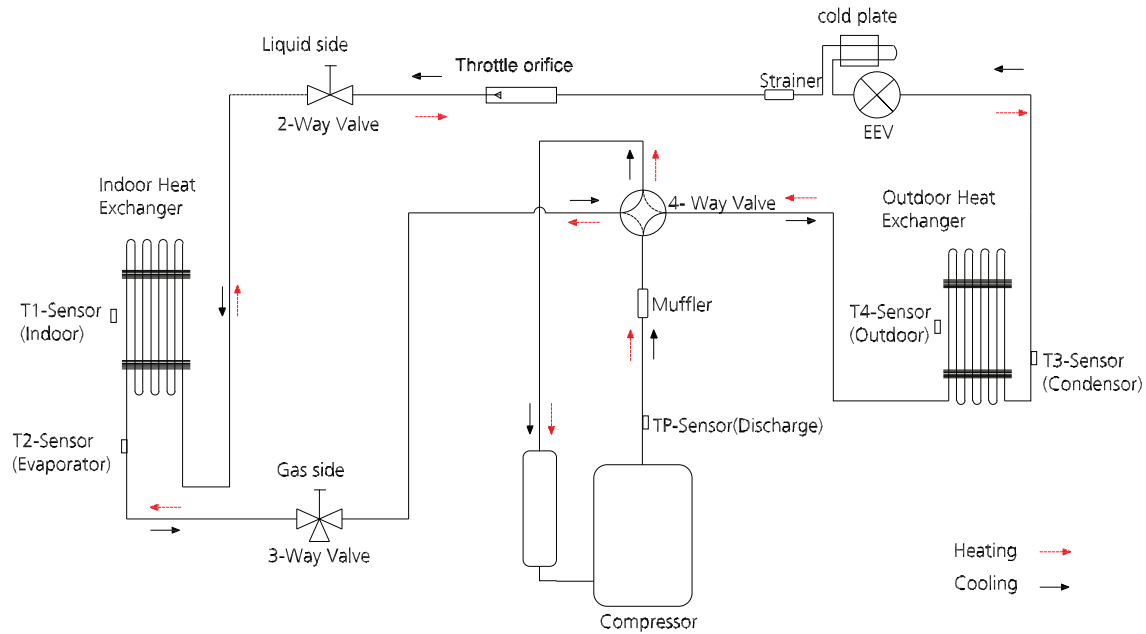


**Fig. 6 — Refrigerant Cycle Diagram - Sizes 9K, 12K**

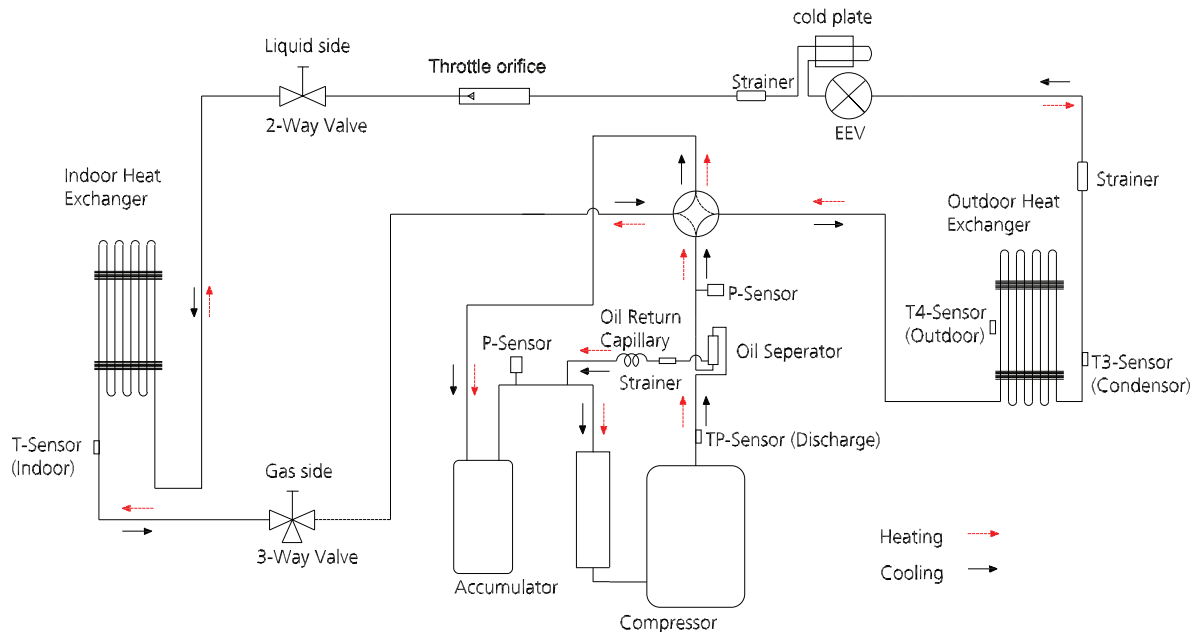


**Fig. 7 — Refrigerant Cycle Diagram - Size 18K**

## REFRIGERANT DIAGRAMS (CONT)



**Fig. 8 — Refrigerant Cycle Diagram - Size 24K**



**Fig. 9 — Refrigerant Cycle Diagram - Sizes 36K, 48K, 58K**

### Refrigerant Lines

#### General refrigerant line sizing:

1. The outdoor units are shipped with a full charge of R410A refrigerant. All charges, line sizing, and capacities are based on runs of 25 ft. (7.6 m). For runs over 25 ft. (7.6 m), consult the unit's product data.
2. Minimum refrigerant line length between the indoor and outdoor units is 10 ft. (3 m).
3. Refrigerant lines should not be buried in the ground. If it is necessary to bury the lines, not more than 36 in. (914 mm) should be buried. Provide a minimum 6 in. (152 mm) vertical rise to the service valves to prevent refrigerant migration.
4. Both lines must be insulated. Use a minimum of 1/2 in. (12.7 mm) thick insulation. Closed-cell insulation is recommended in all long-line applications.
5. Special consideration should be given to isolating the interconnecting tubing from the building structure. Isolate the tubing so vibration or noise is not transmitted into the structure.
6. For piping runs greater than 25 ft. (7.6 m), add refrigerant up to the allowable length as specified in the product data.

## SYSTEM EVACUATION AND CHARGING



### CAUTION

#### UNIT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

Never use the system compressor as a vacuum pump.

Refrigerant tubes and indoor coil should be evacuated using the recommended deep vacuum method of 500 microns. The alternate triple evacuation method may be used if the following procedure is followed. Always break a vacuum with dry nitrogen.

### System Vacuum and Charge

#### Using Vacuum Pump

1. Completely tighten the flare nuts (A, B, C, D, E). Fully open all circuits service valves. Connect the manifold gage charge hose to the charge port of the low side Master service valve to evacuate all circuits at the same time (see Fig. 10).
2. Connect the charge hose to the vacuum pump.
3. Fully open the low side of the manifold gage (see Fig. 11).
4. Start the vacuum pump.
5. Evacuate using the triple evacuation method.
6. After the evacuation is complete, fully close the low side of the manifold gage and stop the vacuum pump operation. The factory charge contained in the outdoor unit is good for up to 25 ft. (8 m) of line length.
7. Disconnect the charge hose from the charge connection of the low side service valve.
8. Fully open the service valves (B and A).
9. Securely tighten the service valve caps.

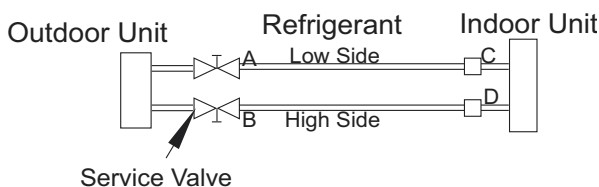


Fig. 10 — Service Valve

A220978

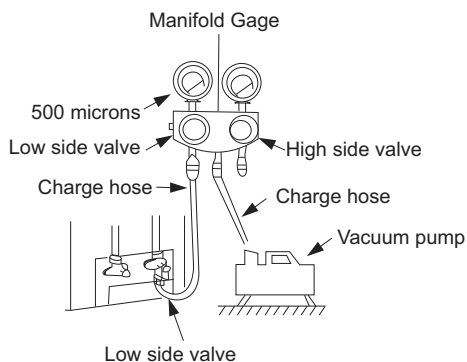


Fig. 11 — Manifold

A220979

#### Deep Vacuum Method

The deep vacuum method requires a vacuum pump capable of pulling a vacuum of 500 microns and a vacuum gage capable of accurately measuring this vacuum depth. The deep vacuum method is the most positive way of ensuring a system is free of air and liquid water (see Fig. 12).

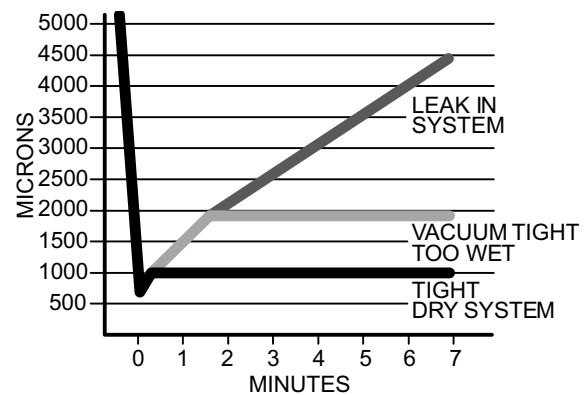


Fig. 12 — Deep Vacuum Method

A220980

#### Triple Evacuation Method

The triple evacuation method should be used. Refer to Fig. 13 and proceed as follows:

1. Pump the system down to 1500 microns and allow the pump to continue operating for an additional 15 minutes.
2. Close the service valves and shut off the vacuum pump.
3. Connect a dry nitrogen cylinder and regulator to the system and break the vacuum until the system reaches 2 psig.
4. Close the service valve and allow the system to stand for one hour. During this time, the dry nitrogen can diffuse throughout the system absorbing moisture.
5. Pump the system down to 1000 microns.
6. Break the vacuum with dry nitrogen (2 psig).
7. Pump the system down to 500 microns.
8. Perform the hold test for 30 minutes.

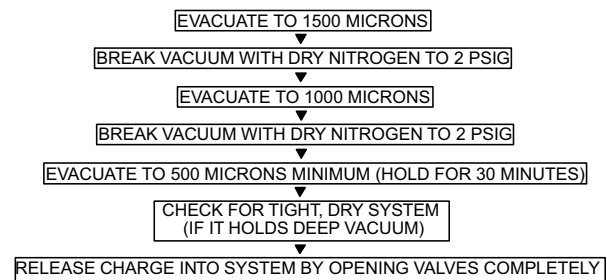


Fig. 13 — Triple Evacuation Method

A220981

#### Final Tubing Check

**IMPORTANT:** Check to ensure the factory tubing on both the indoor and outdoor unit has not shifted during the shipment. Ensure the tubes are not rubbing against each other or any sheet metal. Pay close attention to feeder tubes, ensuring the wire ties on the feeder tubes are secure and tight.

Main Protection  
Fan Speed is Out of Control

When the indoor fan speed remains too low (lower than 300RPM) for 50 seconds, the indoor fan shuts off and restarts 30 seconds later. If the protection occurs three times when the fan motor restarts continuously, the unit stops and the LED displays the failure.

When the outdoor fan speed remains too low (lower than 100 RPM) or too high (higher than 1500 RPM) for 60 seconds, the unit stops and the LED displays the failure then the malfunction clears 30 seconds later.

Evaporator low temperature T2 protection

- $T_2 < 32^{\circ}\text{F}$  ( $0^{\circ}\text{C}$ ), the compressor stops and restarts when  $T_2 \geq 41^{\circ}\text{F}$  ( $5^{\circ}\text{C}$ )
- $32^{\circ}\text{F}$  ( $0^{\circ}\text{C}$ )  $\leq T_2 < 39.2^{\circ}\text{F}$  ( $4^{\circ}\text{C}$ ), the compressor frequency is limited and decreased to the lower level
- $39.2^{\circ}\text{F}$  ( $4^{\circ}\text{C}$ )  $\leq T_2 \leq 44.6^{\circ}\text{F}$  ( $7^{\circ}\text{C}$ ), the compressor maintains the current frequency
- $T_2 > 44.6^{\circ}\text{F}$  ( $7^{\circ}\text{C}$ ), the compressor frequency is not limited

Operation Modes and Functions  
FAN Mode

1. Outdoor fan and compressor stop
2. Temperature setting function is disabled, and no setting temperature appears
3. Indoor fan can be set to high/med/low/auto
4. The louver operates the same in the COOLING mode
5. AUTO fan

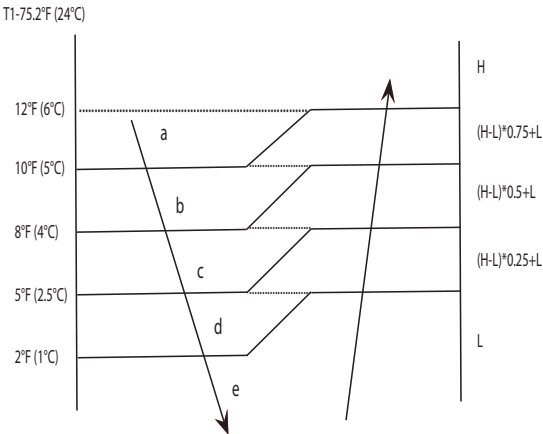


Fig. 14 —Auto Fan

A220982

COOLING Mode

Indoor Fan Running Rules

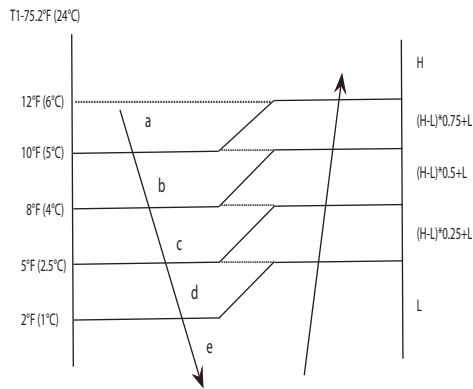
In the COOLING mode, the indoor fan runs constantly and either speed (high, medium, low or auto) can be selected. When the setting temperature is reached, if the compressor stops running, the indoor fan motor runs at the minimum or setting speed. The indoor fan is controlled by the rules shown in Fig. 15.

| Setting Fan Speed | T1-Td °F (°C) | Actual Fan Speed |
|-------------------|---------------|------------------|
| H                 | 8°F (4.5°C)   | H+ (H+=H+G)      |
|                   | 5°F (3.0°C)   | H (=H)           |
|                   | 3°F (1.5°C)   | H- (H-=H-G)      |
| M                 | 8°F (4.5°C)   | M+ (M+=M+Z)      |
|                   | 5°F (3.0°C)   | M (M=M)          |
|                   | 3°F (1.5°C)   | M- (M-=M-Z)      |
| L                 | 8°F (4.5°C)   | L+ (L+=L+D)      |
|                   | 5°F (3.0°C)   | L (L=L)          |
|                   | 3°F (1.5°C)   | L- (L-=L-D)      |

A220983

Fig. 15 — Indoor Fan Running Rules

The **AUTO** fan is controlled by the rules shown in Fig. 16.



A220984

**Fig. 16 — Indoor Fan Running Rules**

### Evaporator Temperature Protection

When the evaporator temperature is lower than the setting value, the compressor stops.

### HEATING Mode

#### Indoor Fan Running Rules

When the compressor is on, the indoor fan can be set to **high/med/low/ auto/mute**. When the indoor unit coil temperature is low, the anti-cold air function starts and the indoor fan motor runs at a low speed and the speed cannot be changed. When the temperature is lower than the setting value, the indoor fan motor stops.

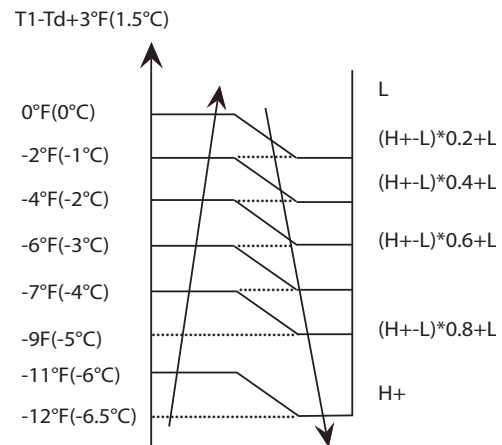
When the indoor temp reaches the setting temperature, the compressor stops and the indoor fan motor runs at the minimum or setting speed (the anti-cold air function is valid). The indoor fan is controlled by the rules shown in Fig. 17.

| Setting Fan Speed | T1-Td + 1.5 °C (3°F) | Actual Fan Speed |
|-------------------|----------------------|------------------|
| H                 | -3°F (-1.5°C)        | H- (H=H-G)       |
|                   | -5°F (-3.0°C)        | H (=H)           |
|                   | -8°F (-4.5°C)        | H+ (H+=H+G)      |
| M                 | -3°F (-1.5°C)        | M- (M-=M-Z)      |
|                   | -5°F (-3.0°C)        | M (M=M)          |
|                   | -8°F (-4.5°C)        | M+ (M+=M+Z)      |
| L                 | -3°F (-1.5°C)        | L- (L-=L-D)      |
|                   | -5°F (-3.0°C)        | L (L=L)          |
|                   | -8°F (-4.5°C)        | L+ (L+=L+D)      |

A220985

**Fig. 17 — Indoor Fan Running Rules**

### Auto Fan Action in HEATING Mode



A220986

**Fig. 18 — Auto Fan Action in HEATING Mode**

### DEFROST Mode

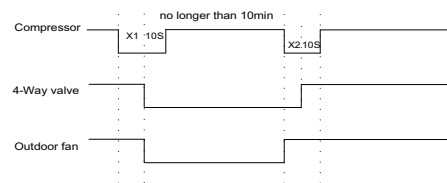
If any one of the following items is satisfied, the unit enters the **DEFROST** mode. After the compressor starts and runs, mark the minimum value of T3 from the 10th minute to the 15th minute as T30.

1. If the compressor runs for 29 minutes and  $T3 < TCDI1$ ,  $T3 + T30SUBT3ONE \leq T30$ .
2. If the compressor runs for 35 minutes and  $T3 < TCDI2$ ,  $T3 + T30SUBT3TWO \leq T30$ .
3. If the compressor runs for 29 minutes. and  $T3 < TCDI3$  for three minutes.
4. If the compressor runs for 120 minutes and  $T3 < 5^\circ\text{F} (-15^\circ\text{C})$ .

### Defrost End

If any one of the following items is satisfied, the **DEFROST** mode ends and the machine reverts to the normal **HEATING** mode.

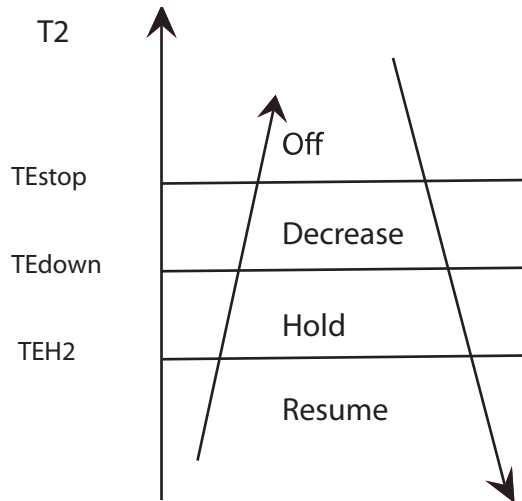
- T3 increases to a point higher than TCDE1.
- T3 maintains a point higher than TCDE2 for 80 seconds.
- Unit runs for 10 minutes in **DEFROST** mode.



A220987

**Fig. 19 — Defrost Action**

## Evaporator Coil Temperature Protection



A220988

**Fig. 20 — Evaporator Coil Temperature Protection**

### NOTE:

- **Off** – Compressor stops
- **Decrease** – Decrease running frequency to lower level
- **Hold** – Maintain the current frequency
- **Resume** – No frequency limit

When the evaporator temperature is higher than the setting protection value, the compressor stops.

### Auto-Mode

This mode can be chosen with the remote control and the setting temperature can be changed between 62°F (17°C)~86°F (30°C).

In the **AUTO** mode, the unit chooses the **COOLING**, **HEATING** or **FAN-ONLY** mode according to DT ( $DT = T1 - Ts$ ).

**Table 3 — Auto Mode**

| $\Delta T = T1 - Ts$                                                               | Running Mode |
|------------------------------------------------------------------------------------|--------------|
| $\Delta T > 4^{\circ}\text{F} (2^{\circ}\text{C})$                                 | Cooling      |
| $-4 (-2^{\circ}\text{C}) \leq \Delta T \leq 2^{\circ}\text{C} (4^{\circ}\text{F})$ | Fan-Only     |
| $\Delta T < -2^{\circ}\text{C} (4^{\circ}\text{F})$                                | Heating      |

The indoor fan runs under **AUTO** fan in the relevant mode. The louver operates the same as in the relevant mode. If the unit switches between the **HEATING** and **COOLING** modes, the compressor stops for 15 minutes and then chooses the mode according to  $T1 - Ts$ . If the setting temperature is modified, the unit chooses the running function again.

### DRYING Mode

#### Indoor Fan Speed is Fixed

The indoor fan speed is fixed at **BREEZE** and cannot be changed. The louver angle is the same as in the **COOLING** mode.

#### Low Indoor Room Temperature Protection

In the **DRYING** mode, if the room temperature is lower than 50°F (10°C), the compressor stops and will not resume until the room temperature exceeds 53.6°F (12°C).

## Timer Function

Timing range is 24 hours

- **TIMER ON:** The unit turns **on** automatically when reaching the setting time.
- **TIMER OFF:** The unit turns **off** automatically when reaching the setting time.
- **TIMER ON/OFF:** The unit turns on automatically when reaching the setting “on” time, and then turns off automatically when reaching the setting “off” time.
- **TIMER OFF/ON:** The unit turns off automatically when reaching the setting “off” time, and then turns on automatically when reaching the setting “on” time.

The timer function will not change the unit’s current operation mode. Suppose the unit is off now; it will not start up first after setting the **TIMER OFF** function. When the setting time is reached, the timer **LED** turns off and the unit’s running mode is the same. The setting time is relative time. The unit exits the timer function if it experiences a malfunction.

## Sleep Function

The **SLEEP** function is available in the **COOLING**, **HEATING** or **AUTO** mode.

The operation process in the **SLEEP** mode is as follows:

- When cooling, the setpoint rises 2°F (1°C) (to a maximum of 86°F (30°C)) every one hour. Two (2) hours later, the room temperature stops rising and the indoor fan remains at a low speed.
- When heating, setpoint decreases 2°F (1°C) (to a minimum of 62°F (17°C)) every one hour. Two (2) hours later, the setpoint stops decreasing and the indoor fan is set at the low speed.

(**Anti-Cold** wind function has the priority).

Operation time in the **SLEEP** mode is seven (7) hours. After seven (7) hours, the unit exits the **SLEEP** mode and powers off.

## FORCED OPERATION Function

When the unit is off, press **TOUCH** to engage the **FORCED AUTO** mode. Press **TOUCH** again within five seconds to engage the **FORCED COOLING** mode. In the **FORCED AUTO**, **FORCED COOLING** or any other operation mode, press **TOUCH** to turn off the unit.

## FORCED OPERATION Mode

In the **FORCED OPERATION** mode, all the general protections and remote control are available.

## Operation Rules

### FORCED COOLING Mode

The compressor runs at the F2 frequency and the indoor fan runs in the **BREEZE** mode. After running for 30 minutes, the machine enters the **AUTO** mode at the 75.2°F (24°C) setting temperature.

### FORCED AUTO Mode:

The **FORCED AUTO** mode is the same as the normal **AUTO** mode with a 75.2°F (24°C) setting temperature.

### **AUTO–RESTART Function**

The indoor unit is equipped with an **AUTO–RESTART** function, which is carried out through an auto–restart module. In the event of a sudden power failure, the module memorizes the setting conditions prior to the power failure. The unit resumes the previous operation setting (not including the **SWING** function) automatically three (3) minutes after the power returns.

If the memorization condition is the **FORCED COOLING** mode, the unit runs in the **COOLING** mode for 30 minutes and switches to the **AUTO** mode at the 75°F (24°C) setpoint. If the unit is off before the power turns off and the unit is required to start up, the compressor delays the start up for 1 minute before powering on. In other instances, the compressor waits three (3) minutes before restarting.

### **FOLLOW ME**

If the indoor PCB receives the signal, which results from pressing **FOLLOW ME** on the remote controller or wired remote controller, the buzzer emits a sound which indicates the **FOLLOW ME** function is initiated. However, when the indoor PCB receives the signal from the remote controller every three minutes, the buzzer will not respond. When the unit is running with the **FOLLOW ME** function, the PCB controls the unit according to the temperature from the **FOLLOW ME** signal, and the temperature collection function of the room temperature sensor is shielded.

When the **FOLLOW ME** function is available, the PCB controls the unit according to the room temperature from the remote controller and the setting temperature. The PCB changes the mode based on information from the remote controller signal, however it is not affected by the setting temperature.

If the unit is running under the **FOLLOW ME** function and the PCB does not receive a signal from the remote controller for seven minutes or after pressing **FOLLOW ME** again, the **FOLLOW ME** function turns off automatically, and the temperature controls the unit according to the room temperature detected from its own room temperature sensor and setting temperature.

### **Refrigerant Leak Detection**

With this new technology, the display area displays “EC” when the outdoor unit detects a refrigerant leak. This function is only active in the **COOLING** mode. This function helps to prevent the compressor from being damaged by a refrigerant leak or a compressor overload.

- **Open Condition:** When the compressor is running, the T2 coil temperature sensor value has no to very little change.

### **46°F (8°C) Heating**

When the compressor is running, the indoor fan motor runs without the **ANTI-COLD** air function. When the compressor is off, the indoor fan motor is off.

### **Silence Operation**

Press **SILENCE** on the remote control to initiate the **SILENCE** function. When the **SILENCE** function is activated, the compressor running frequency remains lower than **F2** and the indoor unit emits a faint breeze, which reduces the noise to the lowest level and creates a quiet and comfortable space for the user.

### **Drain Pump Control**

#### **HEAT**

A call for heat start the outdoor unit and the indoor blower follow its anti-cold blow configuration. The pump will **NOT** turn on if the level switch remains closed (no condensate present). If the level switch opens (water level rise or switch malfunction), the pump starts in approximately 4 seconds.

If the level switch closes within 3 minutes, the pump continues to run for one minute and then the pump turns off. If the level switch remains open for three minutes, the blower turns off, the EEV closes, and the pump turns off.

#### **EE Alarm**

If the level switch closes after the EE alarm has been posted, the EEV opens, the blower starts, and the pump remains off.

#### **COOL**

A call for cool starts the indoor blower. When the compressor starts, the pump turns on and continues to run until the outdoor unit cycles off (single zone) or indoor unit = thermal OFF (MZ). If the level switch opens (water level rise or switch malfunction), the pump remains on. If the level switch closes within 3 minutes, the pump continues to run and no alarm is posted. If the level switch remains open for 3 minutes, the blower turns off, the EEV closes, and the pump powers off.

#### **EE Alarm**

If the level switch closes after the EE alarm has been posted, the EEV opens, the blower starts, and the pump runs when the compressor starts.

**NOTE: The EE alarm condition in the HEAT or COOL mode only directly impacts the compressor operation of a single zone system.**

The Multi-zone outdoor unit continues to run unless the ducted unit is the only indoor unit operating. A ducted unit on a multi-zone system immediately restarts if the EE alarm clears and the compressor is running to serve other units.

## Point Check Function

Press **LED DISPLAY** or **AIR DIRECTION** to check the next or front item's information. When the air conditioner enters the Information Enquiry status, it displays the code name in two seconds (see Table 4).

**Table 4 — Information Enquiry**

| ENQUIRY INFORMATION                | DISPLAYING CODE | MEANING                            |
|------------------------------------|-----------------|------------------------------------|
| T1                                 | T1              | T1 temp.                           |
| T2                                 | T2              | T2 temp.                           |
| T3                                 | T3              | T3 temp.                           |
| T4                                 | T4              | T4 temp.                           |
| T2B                                | Tb              | T2B temp.                          |
| T5                                 | T5              | T5 temp.                           |
| TH                                 | TH              | TH temp.                           |
| Targeted Frequency                 | FT              | Targeted Frequency                 |
| Actual Frequency                   | Fr              | Actual Frequency                   |
| Indoor fan speed                   | IF              | Indoor fan speed                   |
| Outdoor fan speed                  | OF              | Outdoor fan speed                  |
| EXV opening angle                  | LA              | EXV opening angle                  |
| Compressor continuous running time | CT              | Compressor continuous running time |
| Causes of compressor stop          | ST              | Causes of compressor stop          |
| Reserve                            | A0              |                                    |
| Reserve                            | A1              |                                    |
| Reserve                            | .b0             |                                    |
| Reserve                            | .b1             |                                    |
| Reserve                            | .b2             |                                    |
| Reserve                            | .b3             |                                    |
| Reserve                            | .b4             |                                    |
| Reserve                            | .b5             |                                    |
| Reserve                            | .b6             |                                    |
| Reserve                            | .d1             |                                    |
| Reserve                            | Ac              |                                    |
| Reserve                            | Uo              |                                    |
| Reserve                            | Td              |                                    |

When the unit enters the information enquiry status, the code value appears for 25 seconds (see Table 5 on page 14).

**Table 5 — Point Check**

| ENQUIRY INFORMATION                                                         | DISPLAY VALUE           | MEANING                                                                                           | REMARK                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1,T2,T3,T4,<br>T2B,TP,TH,<br>Targeted<br>Frequency,<br>Actual<br>Frequency | -1F,-1E,-1d,-1c,-1b,-1A | -25,-24,-23,-22,-21,-20                                                                           | 1. The displaying temperature is the actual value.<br>2. The temperature is °C no matter what kind of remote control is used.<br>3. T1,T2,T3,T4,T2B display range:-25~70, TP display range:-20~130.<br>4. Frequency display range: 0~159HZ.<br>5. If the actual value exceeds the range, it displays the maximum value or minimum value. |
|                                                                             | -19—99                  | -19—99                                                                                            |                                                                                                                                                                                                                                                                                                                                          |
|                                                                             | A0,A1,...A9             | 100,101,...109                                                                                    |                                                                                                                                                                                                                                                                                                                                          |
|                                                                             | b0,b1,...b9             | 110,111,...119                                                                                    |                                                                                                                                                                                                                                                                                                                                          |
|                                                                             | c0,c1,...c9             | 120,121,...129                                                                                    |                                                                                                                                                                                                                                                                                                                                          |
|                                                                             | d0,d1,...d9             | 130,131,...139                                                                                    |                                                                                                                                                                                                                                                                                                                                          |
|                                                                             | E0,E1,...E9             | 140,141,...149                                                                                    |                                                                                                                                                                                                                                                                                                                                          |
|                                                                             | F0,F1,...F9             | 150,151,...159                                                                                    |                                                                                                                                                                                                                                                                                                                                          |
| Indoor fan speed<br>/Outdoor fan speed                                      | 0                       | OFF                                                                                               |                                                                                                                                                                                                                                                                                                                                          |
|                                                                             | 1,2,3,4                 | Low speed, Medium speed, High speed, Turbo                                                        | For some big capacity motors                                                                                                                                                                                                                                                                                                             |
|                                                                             | 14-FF                   | Actual fan speed = Display value turns to decimal value and then multiply by 10. The unit is RPM. | For some small capacity motors, the display value is from 14-F (hexadecimal), the corresponding fan speed range is from 200-2550 RPM.                                                                                                                                                                                                    |
| EXV opening angle                                                           | 0-FF                    | Actual EXV opening value = Display value turns to decimal value and then multiply by 2.           |                                                                                                                                                                                                                                                                                                                                          |
| Compressor continuous running time                                          | 0-FF                    | 0-255 minutes                                                                                     | If the actual value exceeds the range, it displays the maximum value or minimum value.                                                                                                                                                                                                                                                   |
| Compressor stop causes                                                      | 0-99                    | For a detailed meaning, please consult with an engineer.                                          | Decimal display                                                                                                                                                                                                                                                                                                                          |
| Reserve                                                                     | 0-FF                    |                                                                                                   |                                                                                                                                                                                                                                                                                                                                          |

## TROUBLESHOOTING

This section provides the required flow charts to troubleshoot problems that may arise.

**NOTE: Information required in the diagnoses can be found either on the wiring diagrams or in the appendix.**

### Required Tools:

The following tools are needed when diagnosing the units:

- **Digital multimeter**
- **Screw drivers (Phillips and straight head)**
- **Needle-nose pliers**
- **Refrigeration gauges**

### Recommended Steps

1. Refer to the diagnostic table, see Table 6 on page 16, and determine the problem at hand.
2. Go to the chart, see “DIAGNOSIS AND SOLUTION” on page 17, and follow the steps in the chart for the selected problem.

For ease of service, systems are equipped with diagnostic code display LED's on both the indoor and outdoor units. The outdoor diagnostic display is on the outdoor unit board and is limited to very few errors.

The indoor diagnostic display is a combination of flashing LED's on the display panel on the front of the unit. If possible always check the diagnostic codes displayed on the indoor unit first. The diagnostic codes for the indoor and outdoor units are listed in the appendix (see “APPENDICES” on page 55).

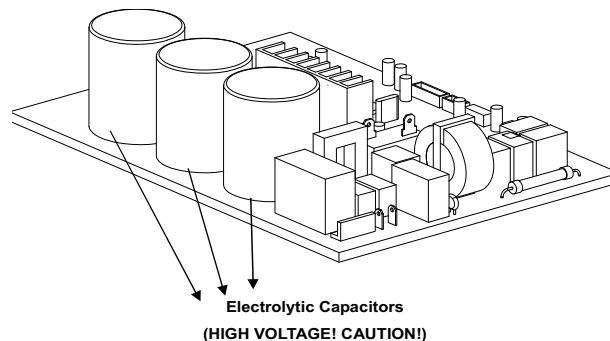
### Safety

Electricity remains in the capacitors even when the power supply is shut off. Do not forget to discharge the electricity in the capacitor.

Problems may occur that are not covered by a diagnostic code, however are covered by the diagnostic flow charts. These problems are typical air conditioning mechanical or electrical issues that can be corrected using standard air conditioning repair techniques.

For problems requiring measurements at the control boards, note the following:

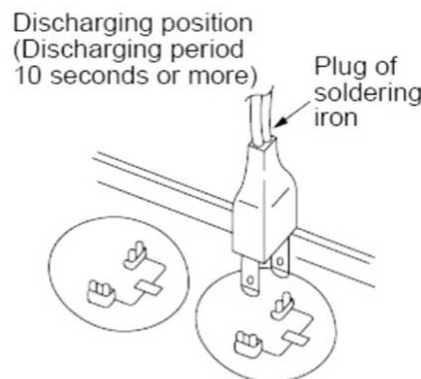
1. Always disconnect the main power.
2. When possible, check the outdoor board first.
3. Start by removing the outdoor unit top cover.
4. Reconnect the main power.
5. Probe the outdoor board inputs and outputs with a digital multimeter referring to the wiring diagrams.
6. Connect the red probe to the hot signal and the black probe to the ground or negative.
7. Note that some of the DC voltage signals are pulsating voltages for signal. This pulse should be rapidly moving at all times when there is a signal present.
8. If it is necessary to check the indoor unit board, disconnect the main power first.
9. Next, remove the unit's front cover and then the control box cover.
10. Carefully remove the indoor board from the control box, place it face up on a plastic surface (not metal).
11. Reconnect the main power and repeat steps 5, 6, and 7.
12. Disconnect the main power before reinstalling the board to avoid a shock hazard and board damage.



**Fig. 21 — Capacitors**

A220989

For other models, connect discharge resistance (approx. 100Ω 40W) or soldering iron (plug) between +, - terminals of the electrolytic capacitor on the contrary side of the outdoor PCB.



**Fig. 22 — Discharging Position**

A220990

**NOTE: Figure 22 is for reference only.**

## Indoor Unit Diagnostic Guides

**Table 6 — Indoor Unit Error Display**

| OPERATION LAMP | TIMER LAMP | DISPLAY | ERROR INFORMATION                                                                                                     | SOLUTION |
|----------------|------------|---------|-----------------------------------------------------------------------------------------------------------------------|----------|
| 1 time         | OFF        | E0      | Indoor unit EEPROM parameter error                                                                                    | Page 17  |
| 2 times        | OFF        | E1      | Indoor / outdoor unit communication error                                                                             | Page 18  |
| 4 times        | OFF        | E3      | The indoor fan speed is operating outside of the normal range                                                         | Page 21  |
| 5 times        | OFF        | E4      | Indoor room temperature sensor T1 is in open circuit or has short circuited                                           | Page 25  |
| 6 times        | OFF        | E5      | Evaporator coil temperature sensor T2 is in open circuit or has short circuited                                       | Page 21  |
| 7 times        | OFF        | EC      | Refrigerant Leakage Detection                                                                                         | Page 26  |
| 8 times        | OFF        | EE      | Water-level alarm malfunction                                                                                         | Page 27  |
| 9 times        | OFF        | E8      | Communication error between master and slave unit (for twins system)                                                  | Page 29  |
| 10 times       | OFF        | E9      | Another indoor unit malfunction (for twins system)                                                                    | --       |
| 11 times       | OFF        | Ed      | Outdoor unit malfunction                                                                                              | Page 28  |
| 1 time         | ON         | F0      | Current overload protection                                                                                           | Page 30  |
| 2 times        | ON         | F1      | Outdoor room temperature sensor T4 is in open circuit or has short circuited                                          | Page 25  |
| 3 times        | ON         | F2      | Condenser coil temperature sensor T3 is in open circuit or has short circuited                                        | Page 25  |
| 4 times        | ON         | F3      | Compressor discharge temperature sensor TP is in a open circuit or has short-circuited                                | Page 25  |
| 5 times        | ON         | F4      | Outdoor unit EEPROM parameter error                                                                                   | Page 17  |
| 6 times        | ON         | F5      | The outdoor fan speed is operating outside of the normal range                                                        | Page 21  |
| 7 times        | ON         | F6      | Evaporator coil outlet temperature sensor T2B is in open circuit or has short circuited (for free-match indoor units) | Page 25  |
| 11 times       | ON         | FA      | Communication error between indoor two chips                                                                          | Page 17  |
| 1 time         | FLASH      | P0      | IPM malfunction or IGBT over-strong current protection                                                                | Page 32  |
| 2 times        | FLASH      | P1      | Over voltage or over low voltage protection                                                                           | Page 33  |
| 3 times        | FLASH      | P2      | Top temperature protection of compressor                                                                              | Page 34  |
| 5 times        | FLASH      | P4      | Inverter compressor drive error                                                                                       | Page 32  |
| 6 times        | FLASH      | P5/--   | Indoor units mode conflict (match with multi outdoor unit)                                                            | Page 43  |
| 7 times        | FLASH      | P6      | High pressure protection or low pressure protection                                                                   | Page 35  |
| 8 times        | FLASH      | P7      | IGBT temperature sensor TH is in open circuit or has short circuited                                                  | Page 25  |

### For other errors:

The display board may show a garbled code or a code not listed in the service manual. Ensure this code is not a temperature reading.

### Troubleshooting:

Test the unit using the remote control. If the unit does not respond to the remote, the indoor PCB needs to be replaced. If the unit responds, the display board needs to be replaced.

**Table 7 — Error Display on Two Way Communication Wired Controller**

| DISPLAY | ERROR INFORMATION                                                                                                                                                                           | SOLUTION |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| F0      | Communication error between wired controller and indoor unit                                                                                                                                | Page 30  |
| EHb3    | Communication error between wire controller and indoor unit (for <b>KSACN0701AAA</b> series wired controller)                                                                               | Page     |
| E1      | Indoor / outdoor unit communication error                                                                                                                                                   | Page 18  |
| E2      | Indoor room temperature sensor T1 is in open circuit or has short circuited                                                                                                                 | --       |
| E3      | Evaporator coil temperature sensor T2 is in open circuit or has short circuited                                                                                                             | Page 21  |
| E4      | Indoor coil outlet temperature sensor T2B is in open circuit or has short circuited                                                                                                         | Page 25  |
| E5      | Outdoor coil temperature sensor malfunction T3 or Outdoor ambient temperature sensor malfunction T4 or Compressor discharge temperature sensor TP is in open circuit or has short circuited | Page 21  |
| E7      | Indoor EEPROM malfunction                                                                                                                                                                   | Page     |
| E8      | The indoor fan speed is operating outside of the normal range                                                                                                                               | Page 29  |
| EA      | Current overload protection                                                                                                                                                                 | Page     |
| E       | IPM malfunction or IGBT over-strong current protection                                                                                                                                      | Page     |
| E       | Outdoor unit malfunction                                                                                                                                                                    | Page     |
| EE      | Water-level alarm malfunction                                                                                                                                                               | Page 27  |
| EF      | Other malfunction                                                                                                                                                                           | --       |

## DIAGNOSIS AND SOLUTION

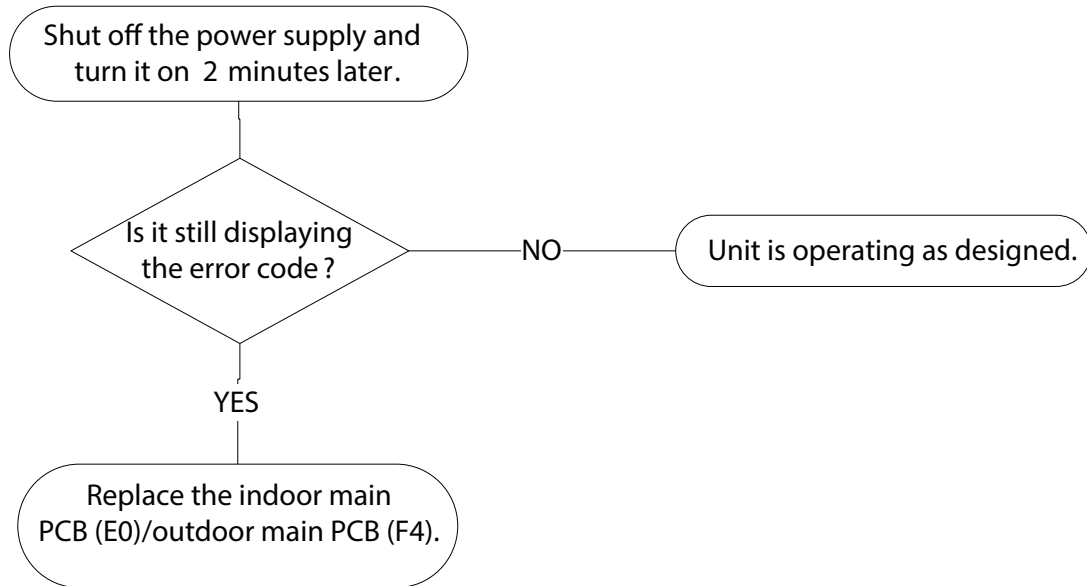
### E0 / F4 (EEPROM parameter error Diagnosis and Solution)

**Description:** Indoor or outdoor PCB main chip does not receive feedback from EEPROM chip.

**Recommended parts to prepare:**

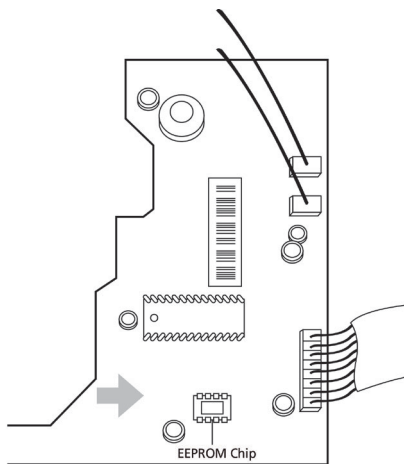
- Indoor PCB
- Outdoor PCB

### Troubleshooting

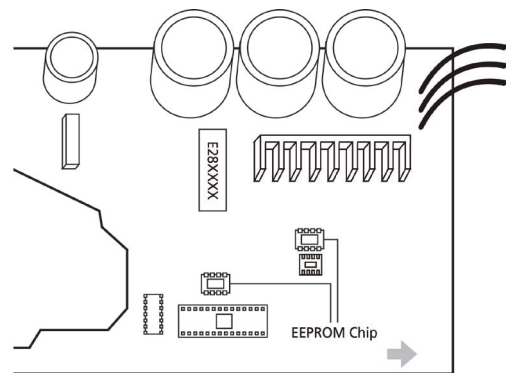


### REMARKS:

**EEPROM:** A read-only memory which contains contents that can be erased and reprogrammed using a pulsed voltage. Review figures 23 and 24 for the location of the EEPROM chip on the indoor and outdoor PCB.



**Fig. 23 — EEPROM Chip (Indoor)**



**Fig. 24 — EEPROM Chip (Outdoor)**

**NOTE:** For certain models, the outdoor PCB cannot be removed separately. In this case, the outdoor electric control box should be replaced as a whole. These images are only for reference and the actual appearance may vary.

## DIAGNOSIS AND SOLUTION (CONT.)

### E1 (Indoor and Outdoor Unit Communication Error Diagnosis and Solution)

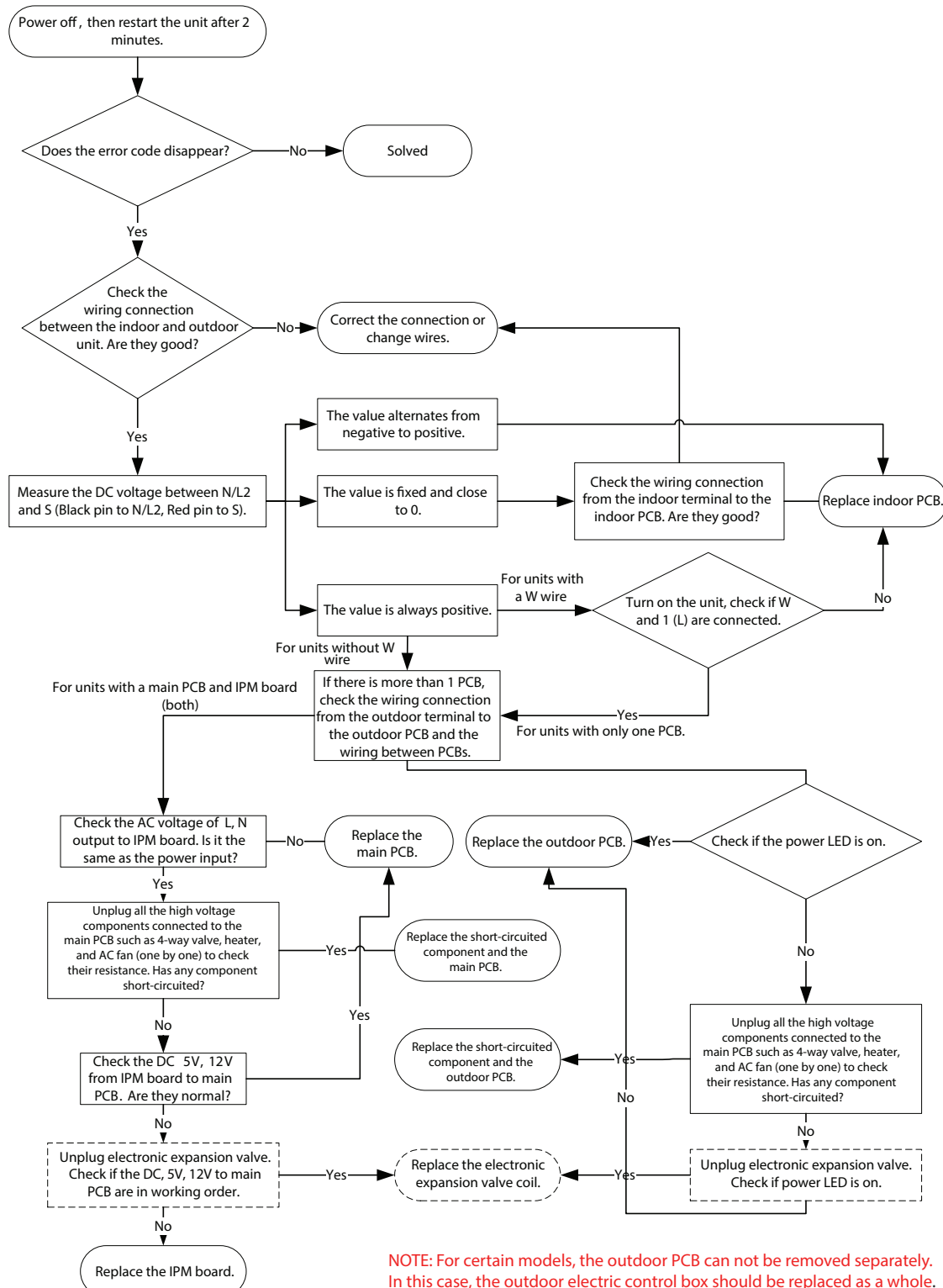
**Description:** Indoor unit cannot communicate with the outdoor unit

**Recommended parts to prepare:**

- Indoor PCB
- Outdoor PCB
- Reactor

#### Troubleshooting

#### For S Communication

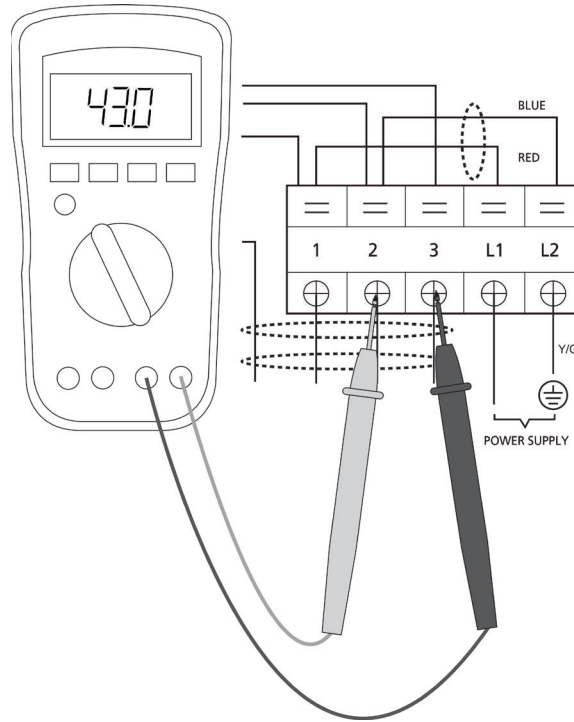


**NOTE:** For certain models, the outdoor PCB can not be removed separately. In this case, the outdoor electric control box should be replaced as a whole.

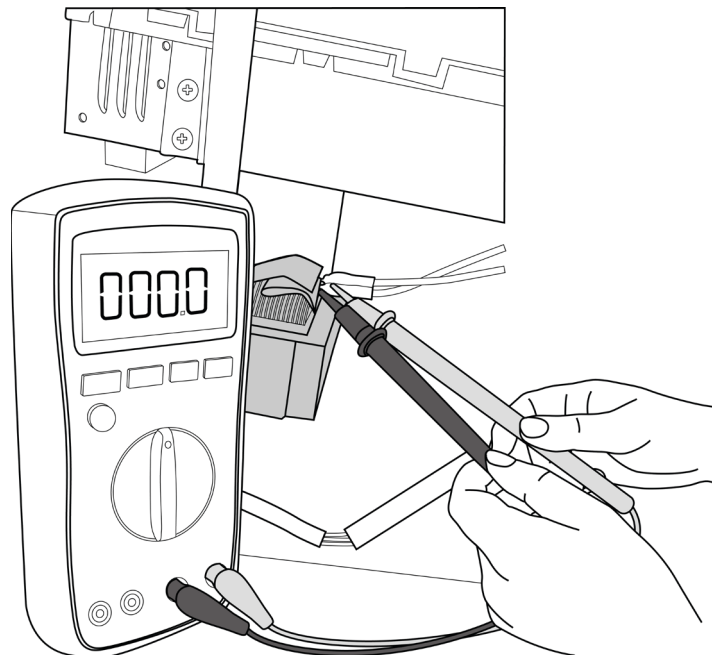
## DIAGNOSIS AND SOLUTION (CONT.)

### Remarks:

- Use a multimeter to test the DC voltage between the 2 port (or S or L2 port) and the 3 port (or N or S port) of the outdoor unit. The red pin of the multimeter connects with the 2 port (or S or L2 port) while the black pin is for the 3 port (or N or S port).
- When the unit is operating normally, the voltage moves alternately between the positive values and negative values.
- If the outdoor unit malfunctions, the voltage was always the positive value.
- Alternately, if the indoor unit malfunctions, the voltage remains at a fixed value.

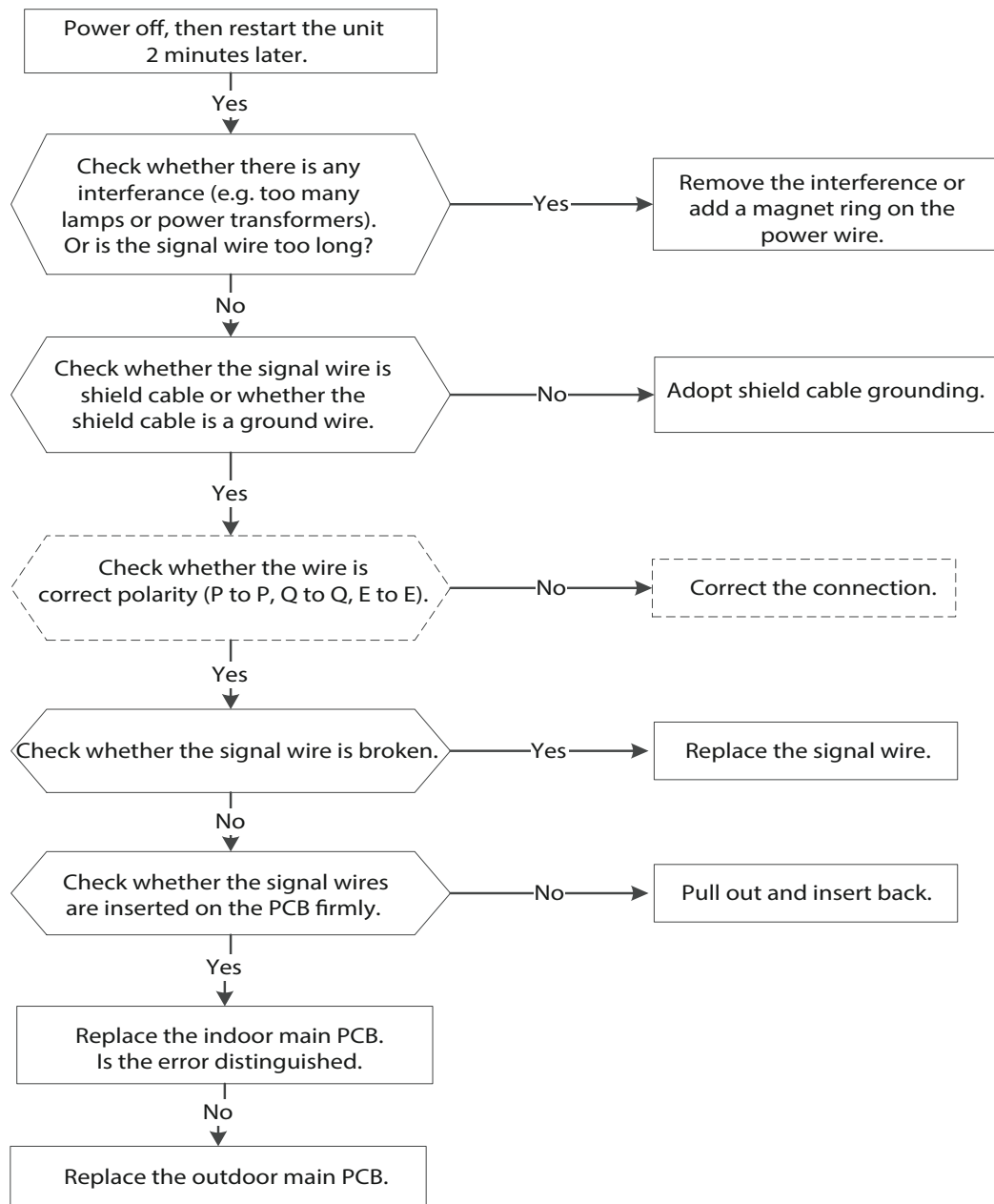


- Use a multimeter to test the reactor resistance which does not connect with the capacitor.
- The normal value should be around zero ohm. Otherwise, the reactor must have malfunctioned.



## DIAGNOSIS AND SOLUTION (CONT.)

### For XYE Communication



## DIAGNOSIS AND SOLUTION (CONT.)

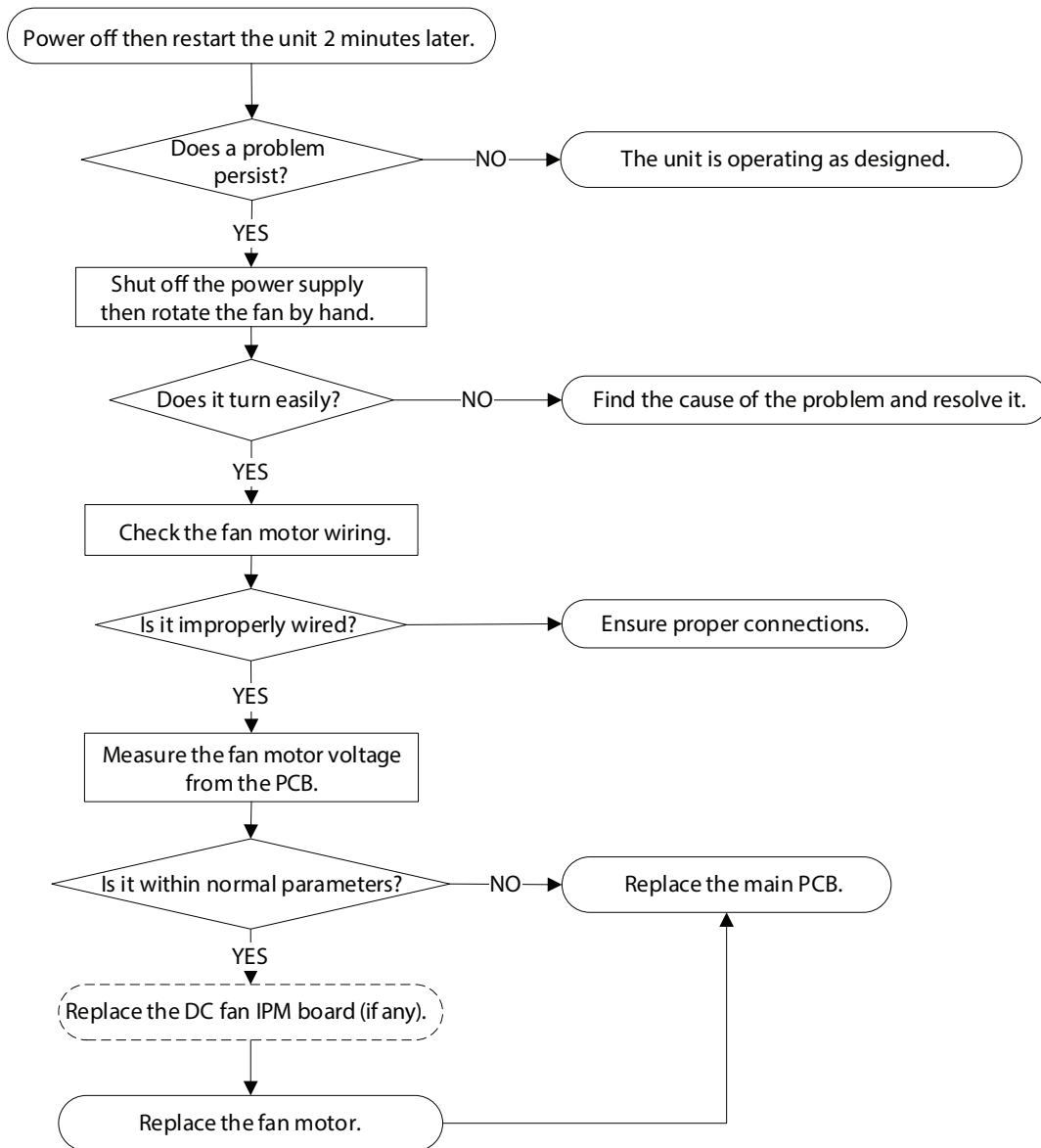
### E3/F5 (Fan Speed Is Operating Outside of Normal Range Diagnosis and Solution)

**Description:** When the indoor / outdoor fan speed remains too slow or too fast for a certain amount of time, the unit ceases operating and the LED displays the failure.

**Recommended parts to prepare:**

- Connection wires
- Fan assembly
- Fan motor
- PCB

#### Troubleshooting



**NOTE:** For certain models, the outdoor PCB cannot be removed separately. In this case, the outdoor electric control box should be replaced as a whole.

## DIAGNOSIS AND SOLUTION (CONT.)

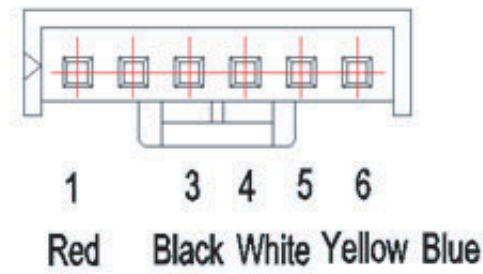
### Index

- Indoor or Outdoor DC Fan Motor (control chip is in fan motor)

Power on the unit and when the unit is in the **STANDBY** mode, measure the voltage of pin1-pin3, pin4-pin3 in the fan motor connector. If the voltage value is not in the range shown in Table 8, the PCB has malfunctioned and needs to be replaced.

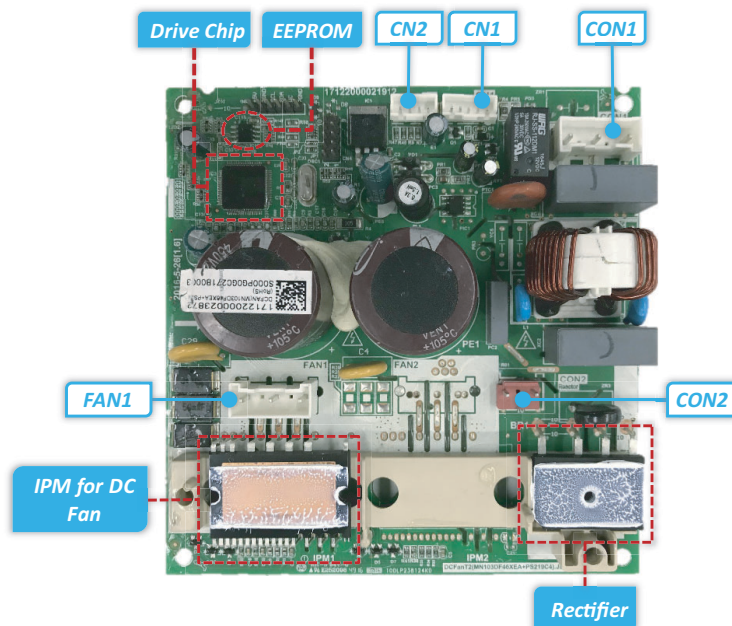
**Table 8 — Signals**

| NO. | COLOR  | SIGNAL | VOLTAGE    |
|-----|--------|--------|------------|
| 1   | Red    | Vs/Vm  | 200V~380V  |
| 2   | ---    | ---    | ---        |
| 3   | Black  | GND    | 0V         |
| 4   | White  | Vcc    | 13.5-16.5V |
| 5   | Yellow | Vsp    | 0~6.5V     |
| 6   | Blue   | FG     | 13.5-16.5V |



- Indoor DC Fan IPM Board (Duct and Ceiling-floor Unit)

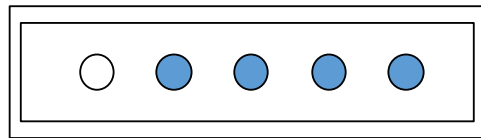
Power on and when the unit is in the **STANDBY** mode, measure the voltage of the CON1, pin1-pin2 and pin3-pin2 of CN1 in the DC motor driver board. If the voltage value is not in the range (see Table 9), the indoor main PCB has an issue and needs to be replaced.



**Table 9 — Signals**

| PORT | DESCRIPTION                 | PARAMETER | REMARK              |
|------|-----------------------------|-----------|---------------------|
| CON1 | Power input for the PCB     | 230V/AC   |                     |
| CN1  | Communication with main PCB | DC        |                     |
| CN2  | Test Port                   | 5V/DC     | For debugging board |
| CN23 | UVW output for DC fan motor |           |                     |
| CON2 | Ports for reactor           |           |                     |

## CN1 Communication with The Main PCB

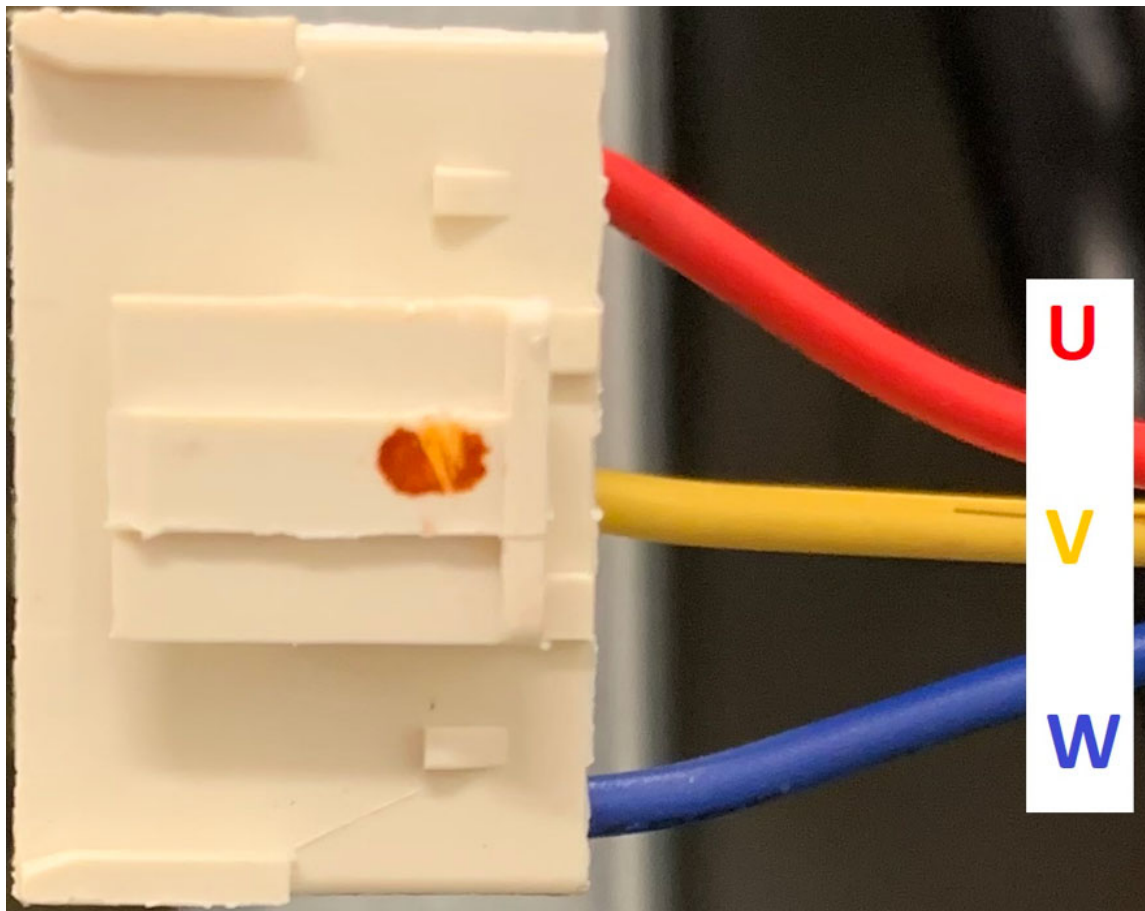


**5 4 3 2 1**

| NO. | SIGNAL | VOLTAGE |
|-----|--------|---------|
| 1   | Vcc    | +15V    |
| 2   | GND    |         |
| 3   | TXD    | 0~6V    |
| 4   | RXD    | 0~15V   |
| 5   | --     | --      |

### 3. Outdoor DC Fan Motor (control chip is in outdoor PCB)

Release the UVW connector. Measure the resistance of U-V, U-W, V-W. If the resistance is not equal to each other, the fan motor has an issue and needs to be replaced. Otherwise the PCB has malfunctioned and needs to be replaced.



4. Outdoor DC Fan IPM Board (for some double fan models)

Power on and when the unit is in the **STANDBY** mode, measure the voltage of CON1, pin1-pin2 and pin3-pin2 of CN1 in the DC motor driver board. If the voltage value is not in the range (see Table 10), the outdoor main PCB has an issue and needs to be replaced.

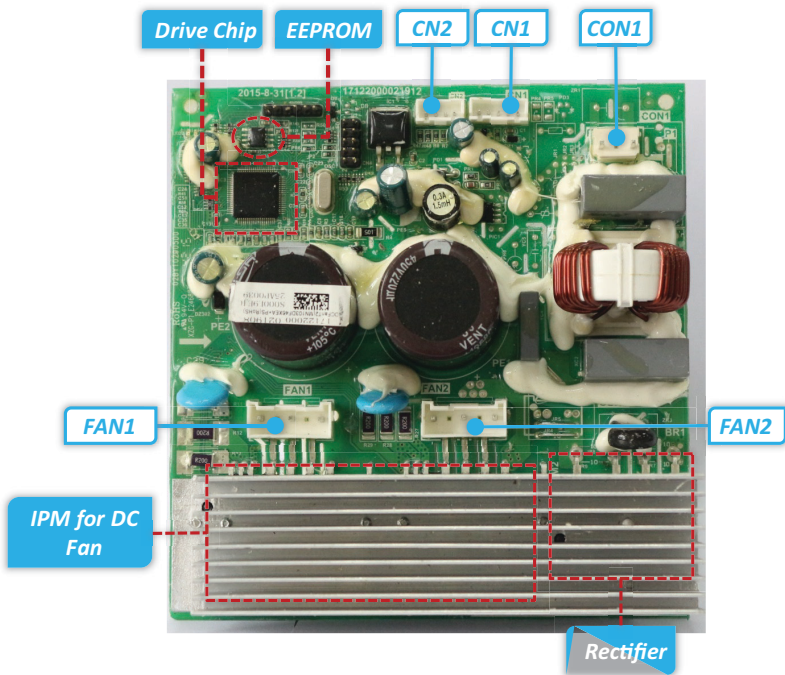
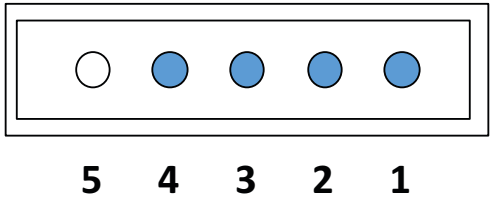


Table 10 — Signals

| PORT | DESCRIPTION                 | PARAMETER   | REMARK              |
|------|-----------------------------|-------------|---------------------|
| CON1 | Power input for the PCB     | 192-380V/DC |                     |
| CN1  | Communication with main PCB | DC          |                     |
| CN2  | Test Port                   | 5V/DC       | For debugging board |
| FAN1 | UVW output for DC fan motor |             |                     |
| FAN2 | UVW output for DC fan motor |             |                     |

CN1 Communication with the Main PCB (CONT)



| NO | SIGNAL | VOLTAGE    |
|----|--------|------------|
| 1  | Vcc    | 13.5-16.5V |
| 2  | GND    | 0V         |
| 3  | VSP    | 0~6V       |
| 4  | FG     | 13.5-16.5V |
| 5  | ----   | ---        |

## DIAGNOSIS AND SOLUTION (CONT.)

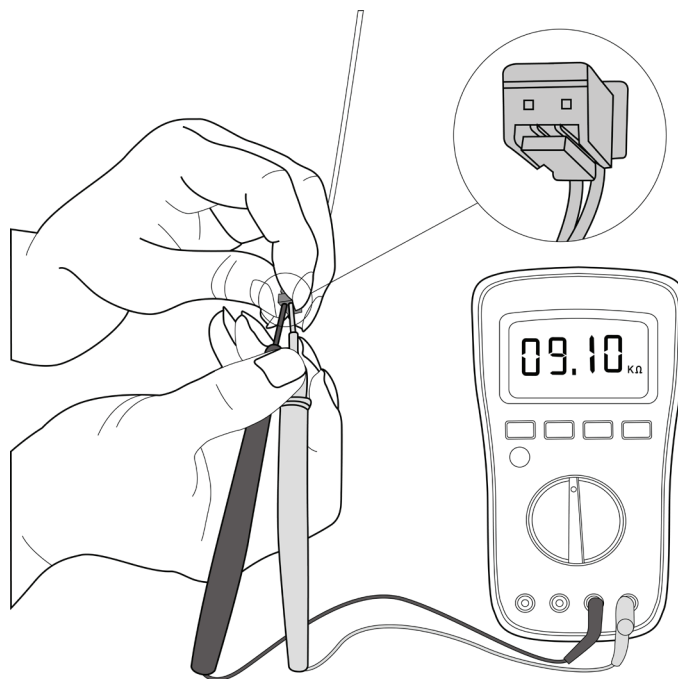
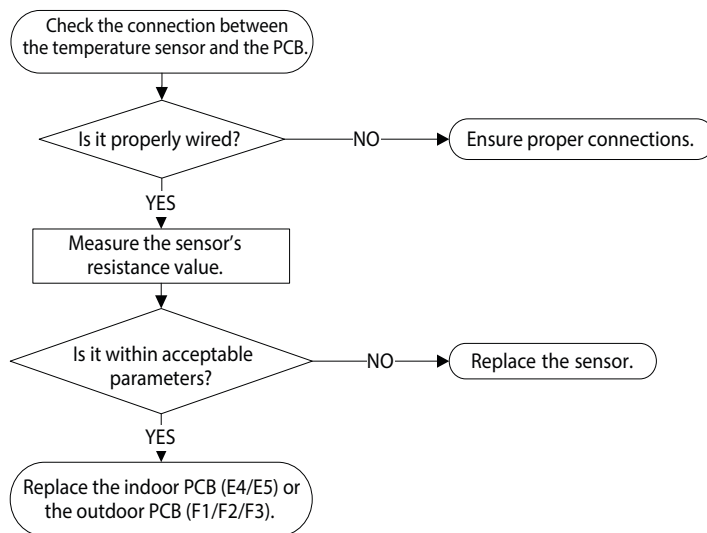
### E4/E5/F1/F2/F3/F6/P7 (Open Circuit or Short Circuit of Temperature Sensor Diagnosis and Solution)

**Description:** If the sampling voltage is lower than 0.06V or higher than 4.94V, the LED displays a failure.

**Recommended parts to prepare:**

- Connection wires
- Sensors
- PCB

#### Troubleshooting



**NOTE:** For certain models, the outdoor PCB cannot be removed separately. In this case, the outdoor electric control box should be replaced as a whole. The image in this manual and the values are only for reference. Actual appearance and values may vary.

## DIAGNOSIS AND SOLUTION (CONT.)

### EC (Refrigerant Leakage Detection Diagnosis and Solution)

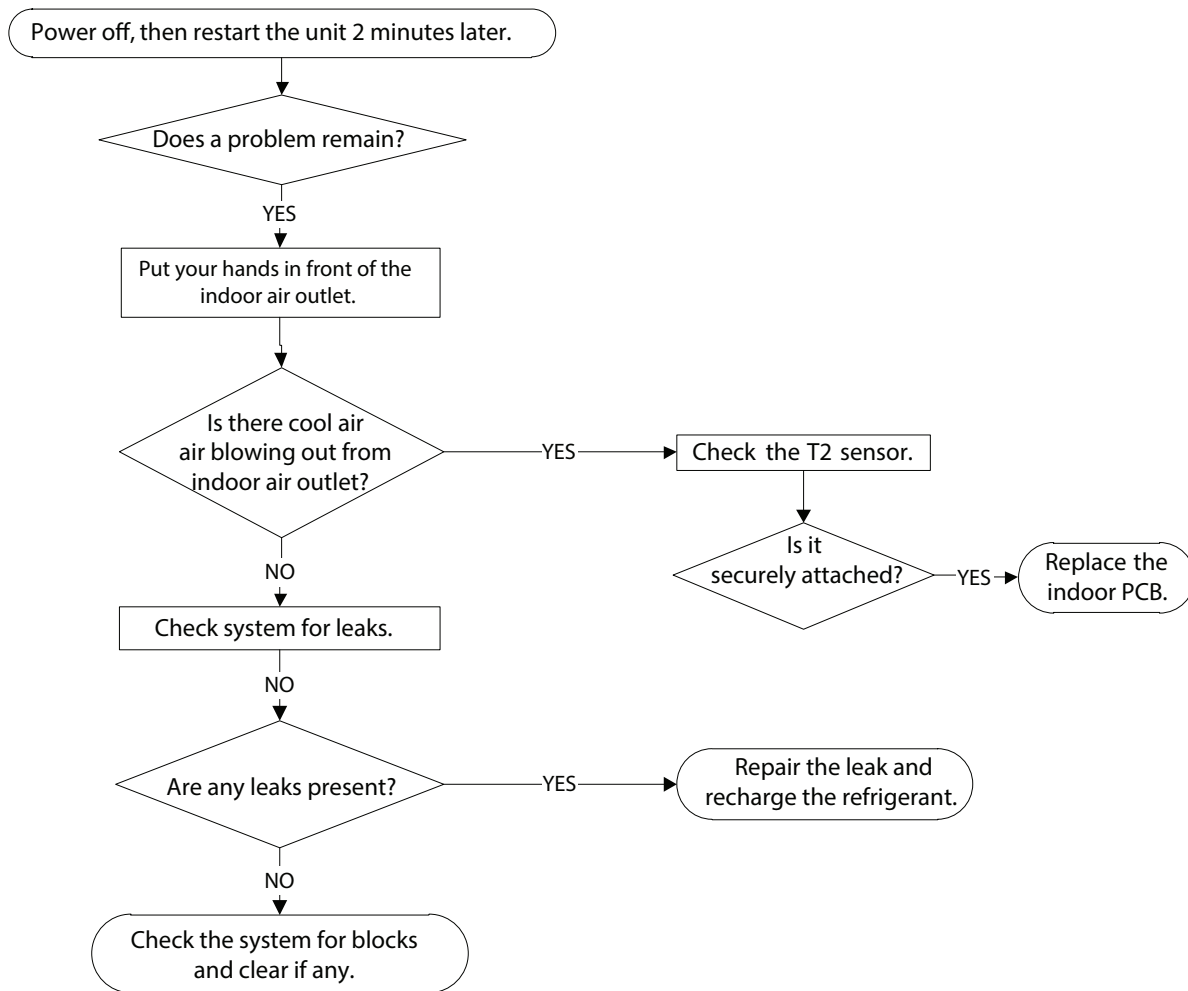
**Description:** Define the compressor's evaporator coil temperature(T2) as it starts running as Tcool.

In the first 5 minutes after the compressor starts up, if  $T2 < T_{cool} - 1^{\circ}\text{C} (1.8^{\circ}\text{F})$  does not maintain for 4 seconds and the compressor running frequency (higher than 50Hz) does not maintain for 3 minutes, and this situation occurs 3 times, the display area displays "EC" and air conditioner turns off.

**Recommended parts to prepare:**

- T2 sensor
- Indoor PCB
- Additional refrigerant

### Troubleshooting



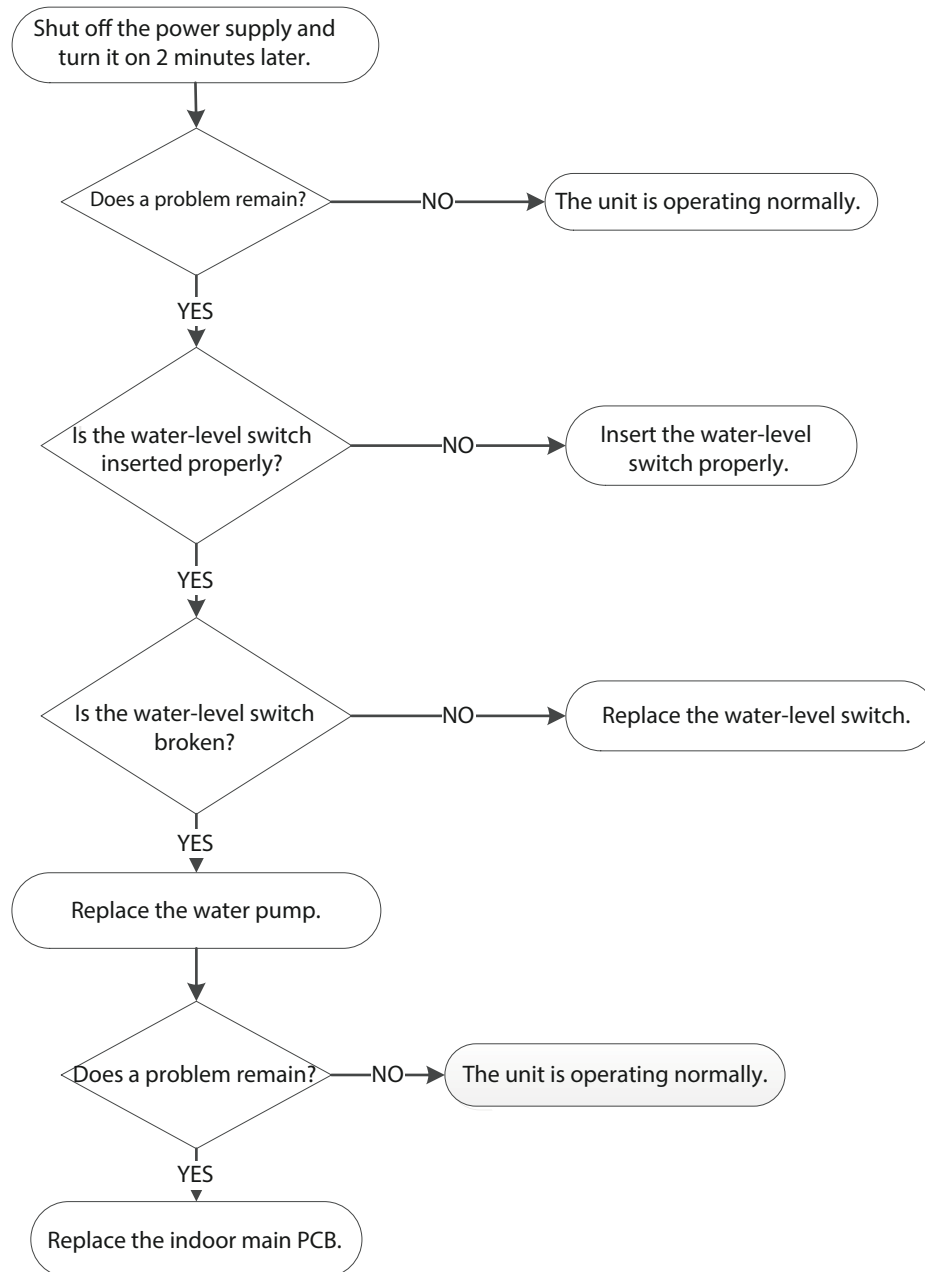
## DIAGNOSIS AND SOLUTION (CONT.)

### EE (Water-Level Alarm Malfunction Diagnosis and Solution)

#### Recommended parts to prepare:

- Connection wires
- Water-level switch
- Water pump
- Indoor PCB

#### Troubleshooting



## DIAGNOSIS AND SOLUTION (CONT.)

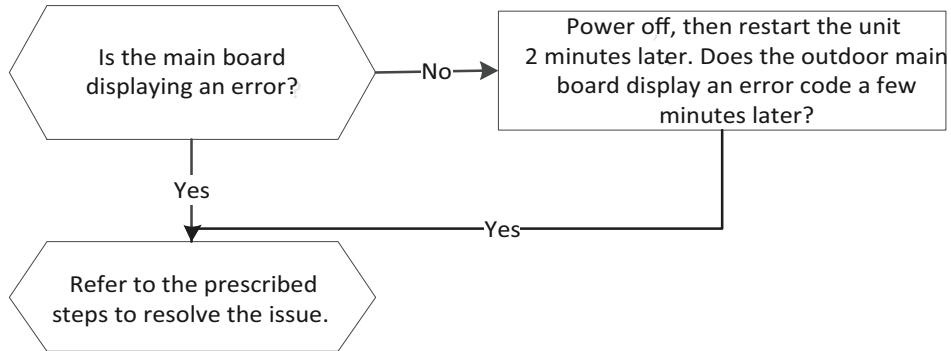
### Ed (Outdoor unit malfunction Diagnosis and Solution)

**Description:** The indoor unit detects an error with the outdoor unit.

**Recommended parts to prepare:**

- Outdoor unit

#### Troubleshooting



## DIAGNOSIS AND SOLUTION (CONT.)

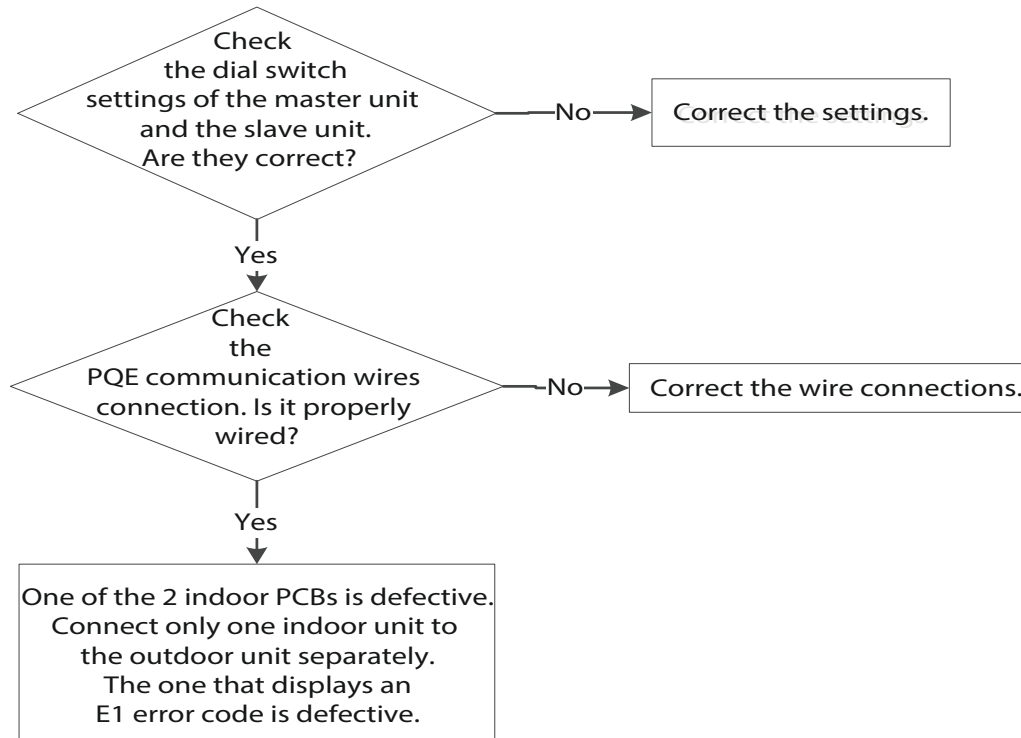
### E8 (Communication error between master and slave unit (for twins system) Diagnosis and Solution)

**Description:** When set in the twins system, the master unit and slave unit cannot be recognized normally.

**Recommended parts to prepare:**

- Connection wires
- Indoor PCB

#### Troubleshooting



## DIAGNOSIS AND SOLUTION (CONT.)

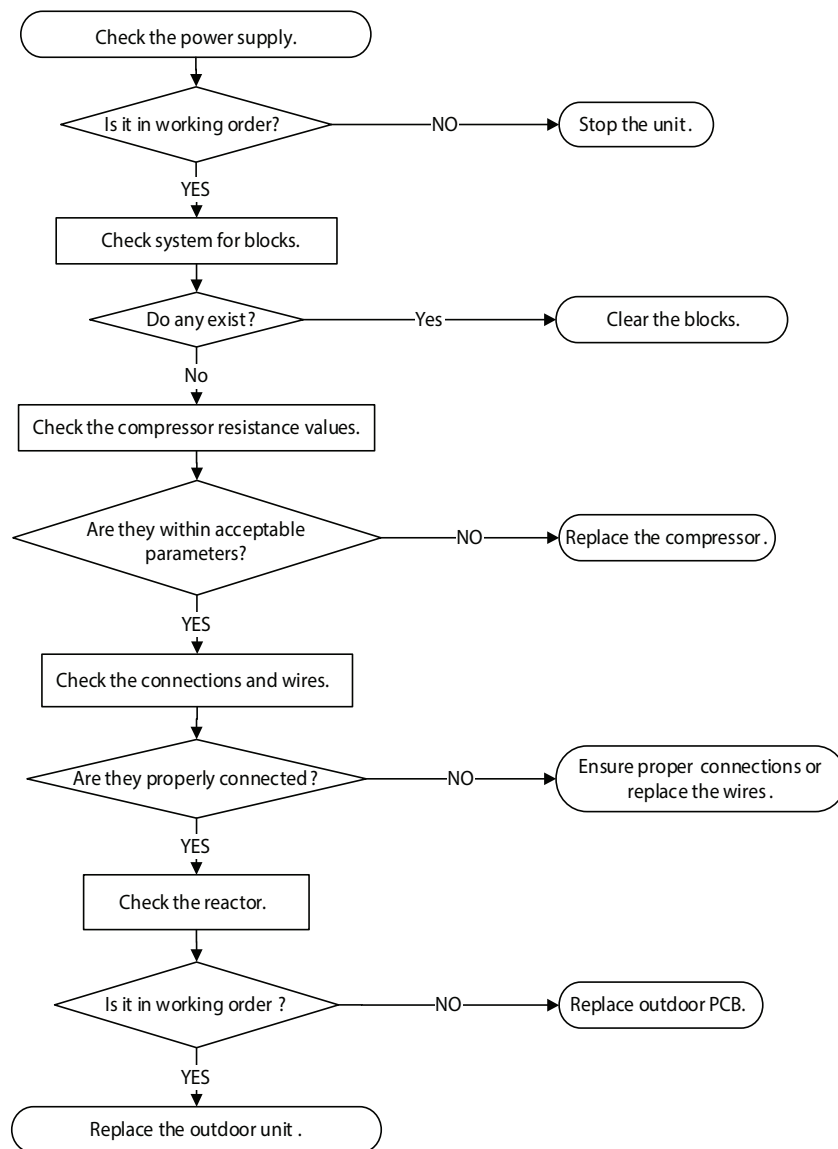
### F0 (Overload current protection Diagnosis and Solution)

**Description:** An abnormal current rise is detected by checking the specified current detection circuit.

**Recommended parts to prepare:**

- Outdoor PCB
- Connection wires
- Compressor
- Reactor

### Troubleshooting



## DIAGNOSIS AND SOLUTION (CONT.)

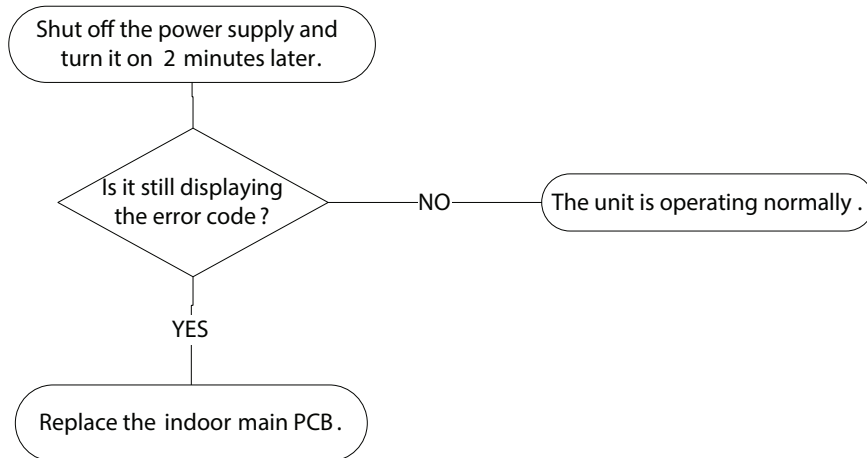
### FA (Communication error between indoor two chips Diagnosis and Solution)

**Description:** Indoor PCB main chip does not receive feedback from another chip.

**Recommended parts to prepare:**

- Indoor PCB

#### Troubleshooting



## DIAGNOSIS AND SOLUTION (CONT.)

### P0 (IPM malfunction Diagnosis and Solution) and P4 (Inverter compressor drive error Diagnosis and Solution)

**Description:** When the voltage signal that the IPM sends to the compressor drive chip is abnormal, the display LED displays “PC 00” and the air conditioner turns off.

#### Recommended parts to prepare:

- Connection wires
- IPM module board
- Outdoor fan assembly
- Compressor
- Outdoor PCB

#### Troubleshooting

At first test, the resistance between every two ports of U, V, W of IPM and P, N. If any result is 0 or close to 0, the IPM is defective. Otherwise, follow this flowchart.



## DIAGNOSIS AND SOLUTION (CONT.)

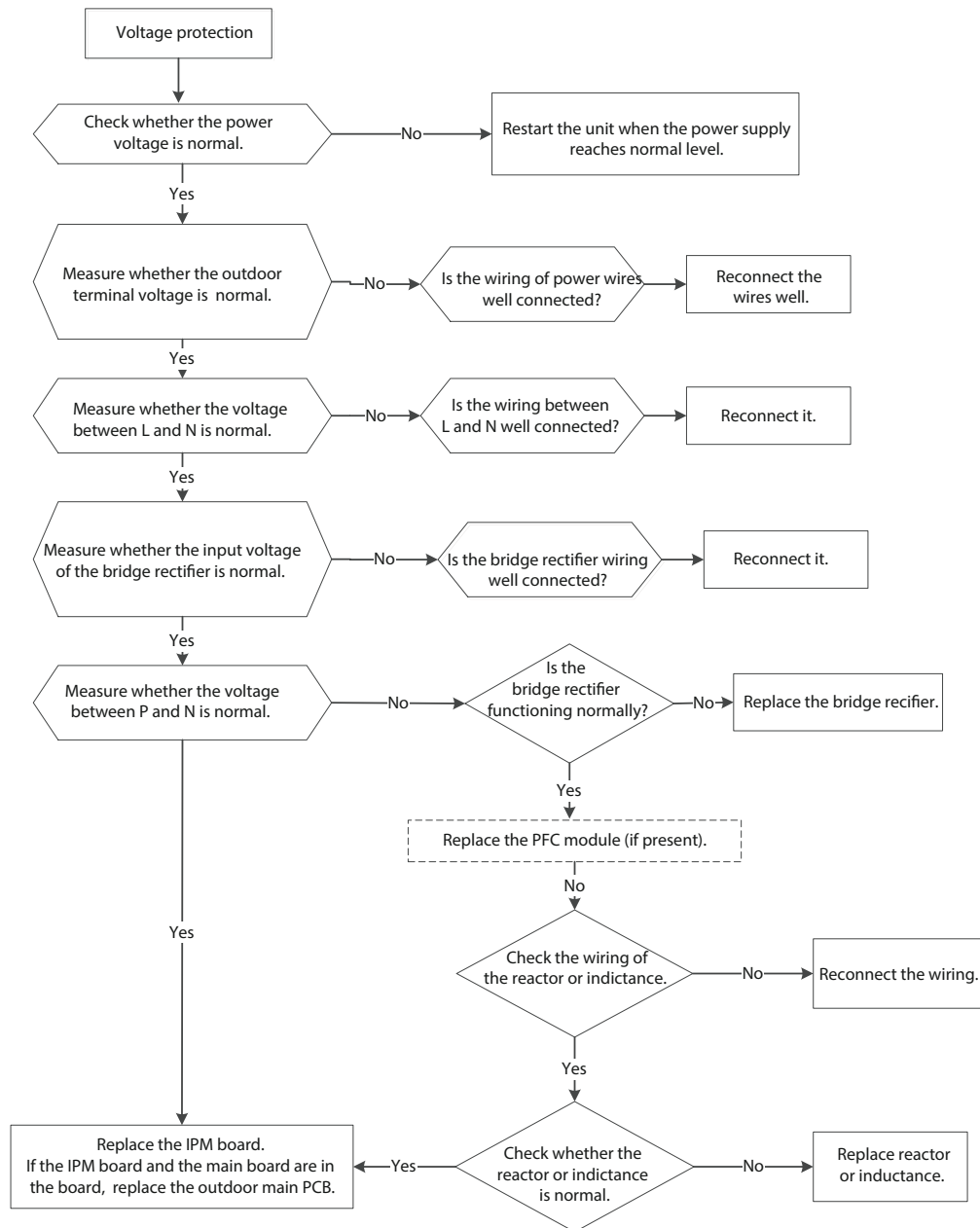
### P1 (Over voltage or too low voltage protection Diagnosis and Solution)

**Description:** Abnormal increases or decreases in voltage are detected by checking the specified voltage detection circuit.

**Recommended parts to prepare:**

- Power supply wires
- IPM module board
- Outdoor PCB
- Bridge rectifier
- PFC circuit or reactor

### Troubleshooting



## DIAGNOSIS AND SOLUTION (CONT.)

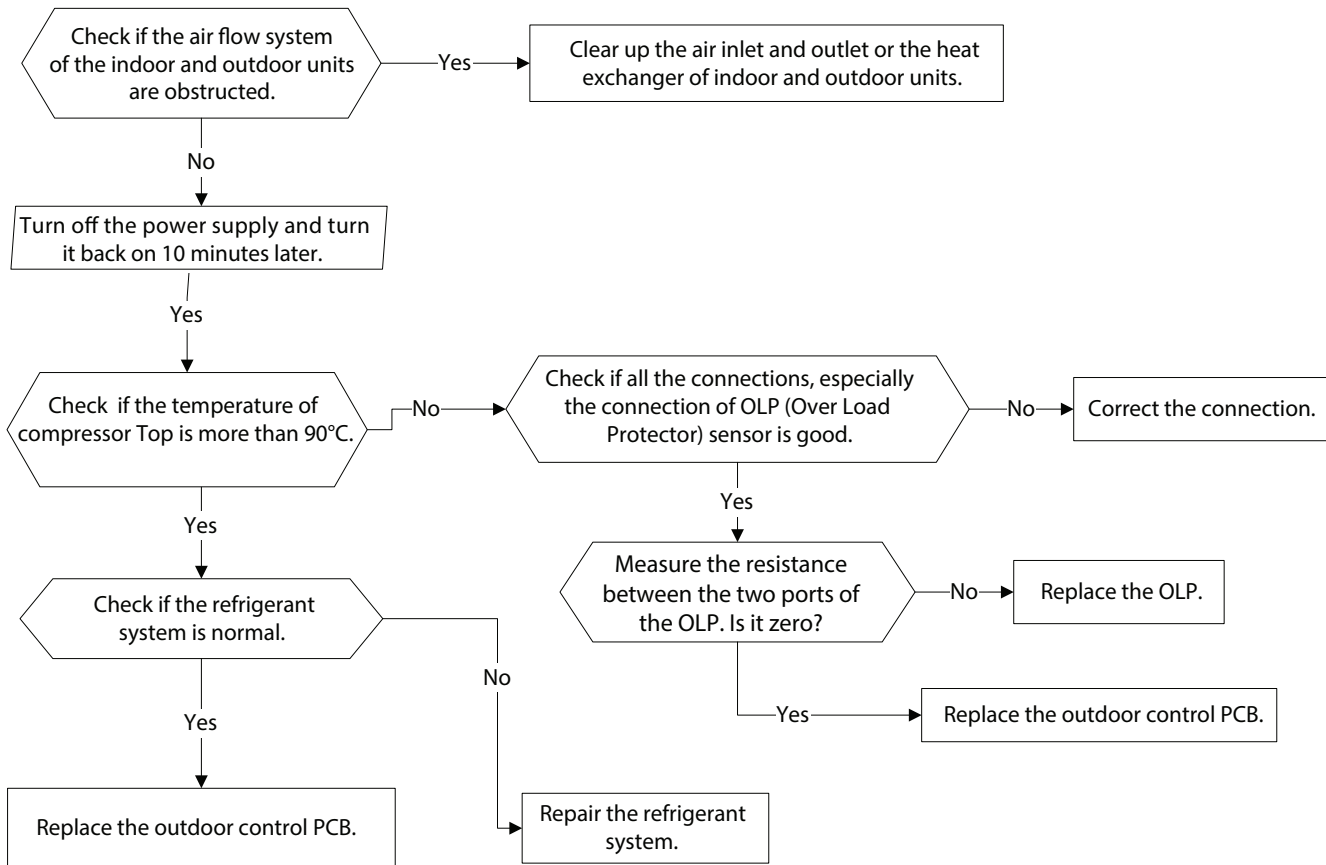
### P2 (Top temperature protection of the compressor Diagnosis and Solution)

**Description:** If the sampling voltage is not 5V, the LED displays a failure.

**Recommended parts to prepare:**

- Connection wires
- Outdoor PCB
- Overload protector

#### Troubleshooting



## DIAGNOSIS AND SOLUTION (CONT.)

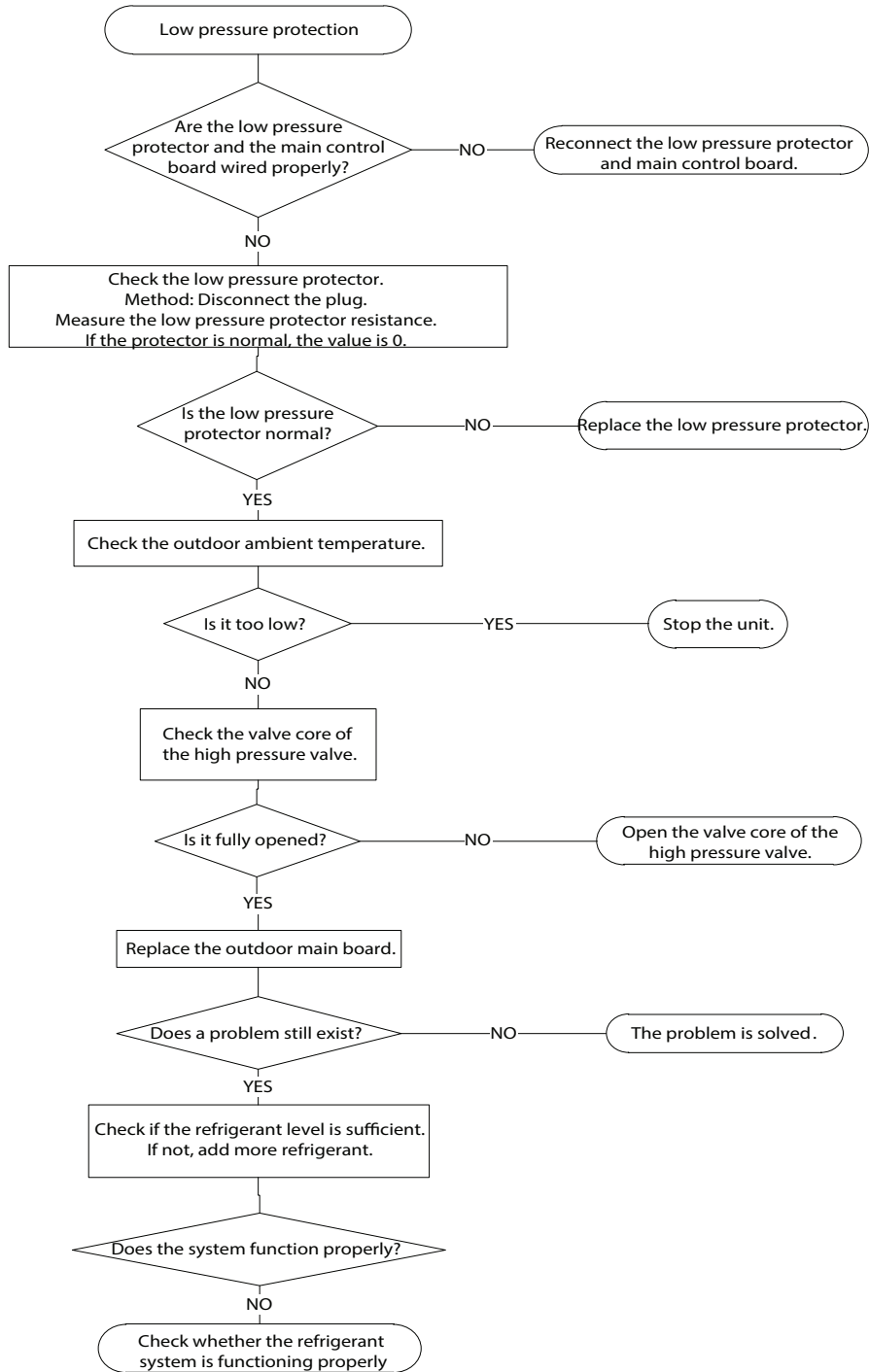
### P6 (Low ambient temperature protection)

**Description:** If the sampling voltage is not 5V, the LED displays a failure code.

**Recommended parts to prepare:**

- Wiring mistake
- Over load protector
- System blockages
- Outdoor PCB

### Troubleshooting



## DIAGNOSIS AND SOLUTION (CONT.)

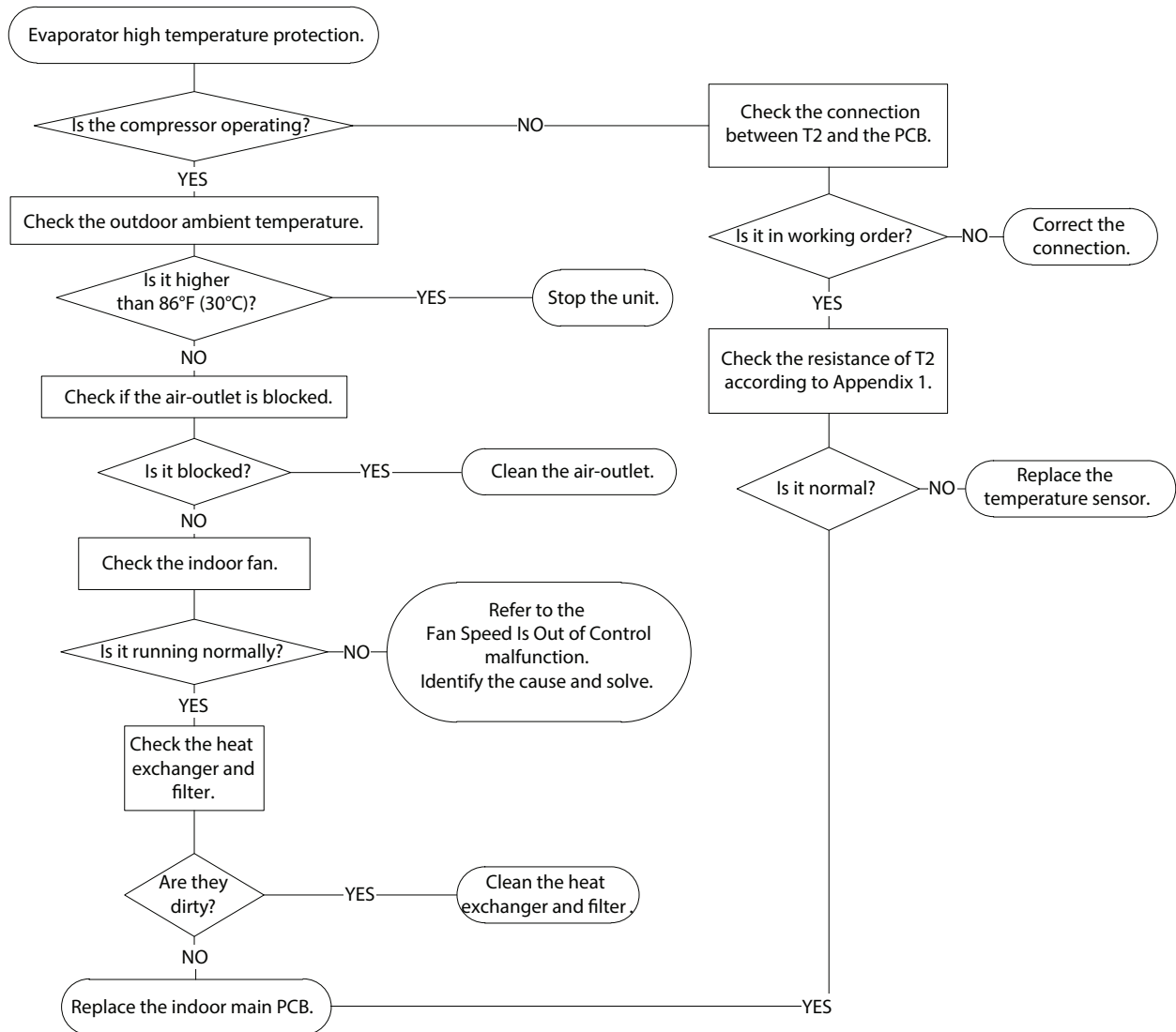
### J0 (Evaporator high temperature protection)

**Description:** When the evaporator coil temperature is more than 140°F(60°C), the unit stops. It starts again only when the evaporator coil temperature is less than a certain value.

#### Recommended parts to prepare:

- Evaporator coil temperature sensor
- Fan
- PCB

#### Troubleshooting



## DIAGNOSIS AND SOLUTION (CONT.)

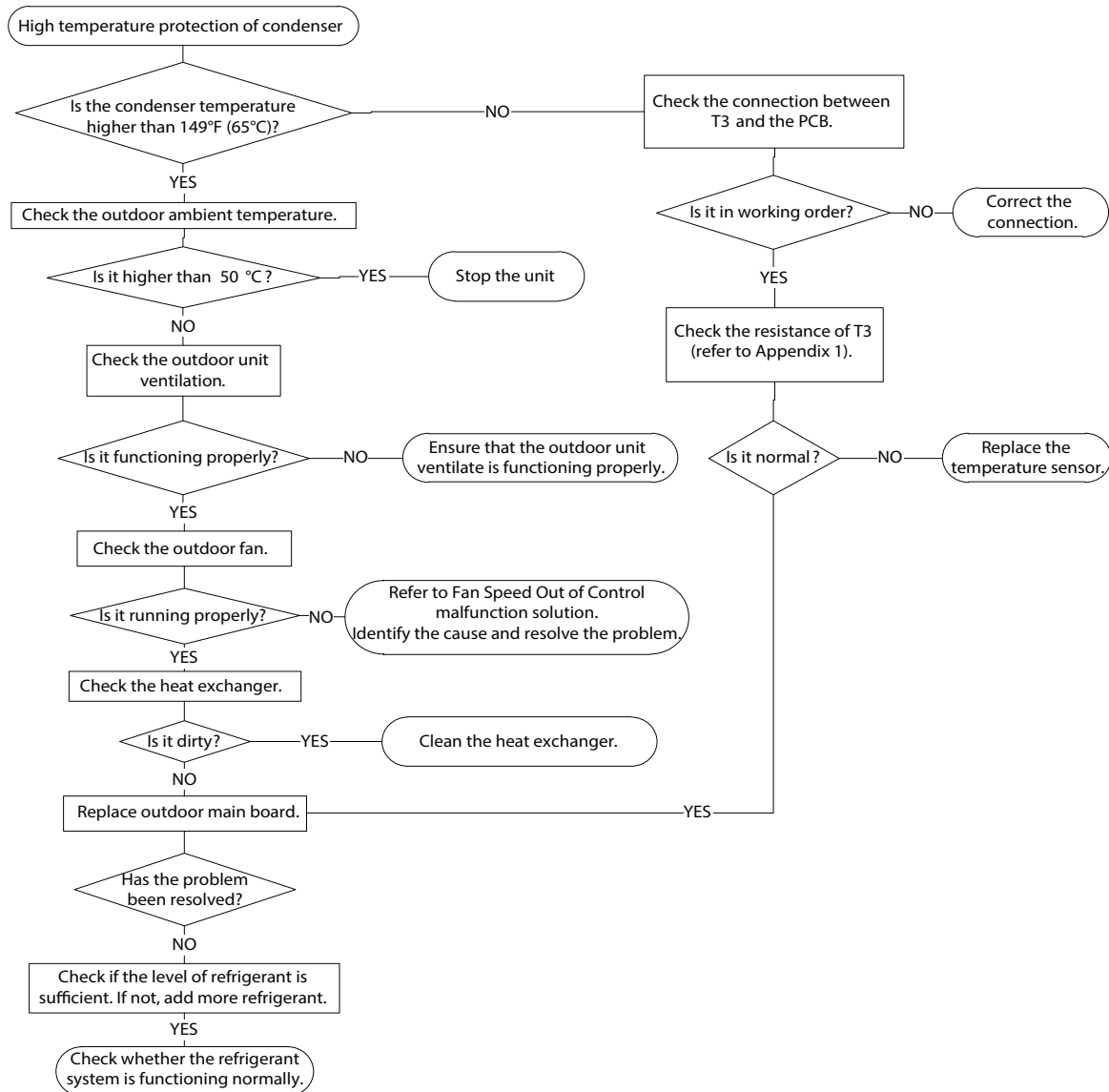
### J1 (Condenser high temperature protection)

**Description:** When the outdoor pipe temperature is higher than  $TP3+5^{\circ}\text{C}$ , the unit stops. The unit starts again only when the outdoor pipe temperature is lower than  $TP3-3^{\circ}\text{C}$ .

#### Recommended parts to prepare:

- Condenser temperature sensor
- System leakage or block

#### Troubleshooting



## DIAGNOSIS AND SOLUTION (CONT.)

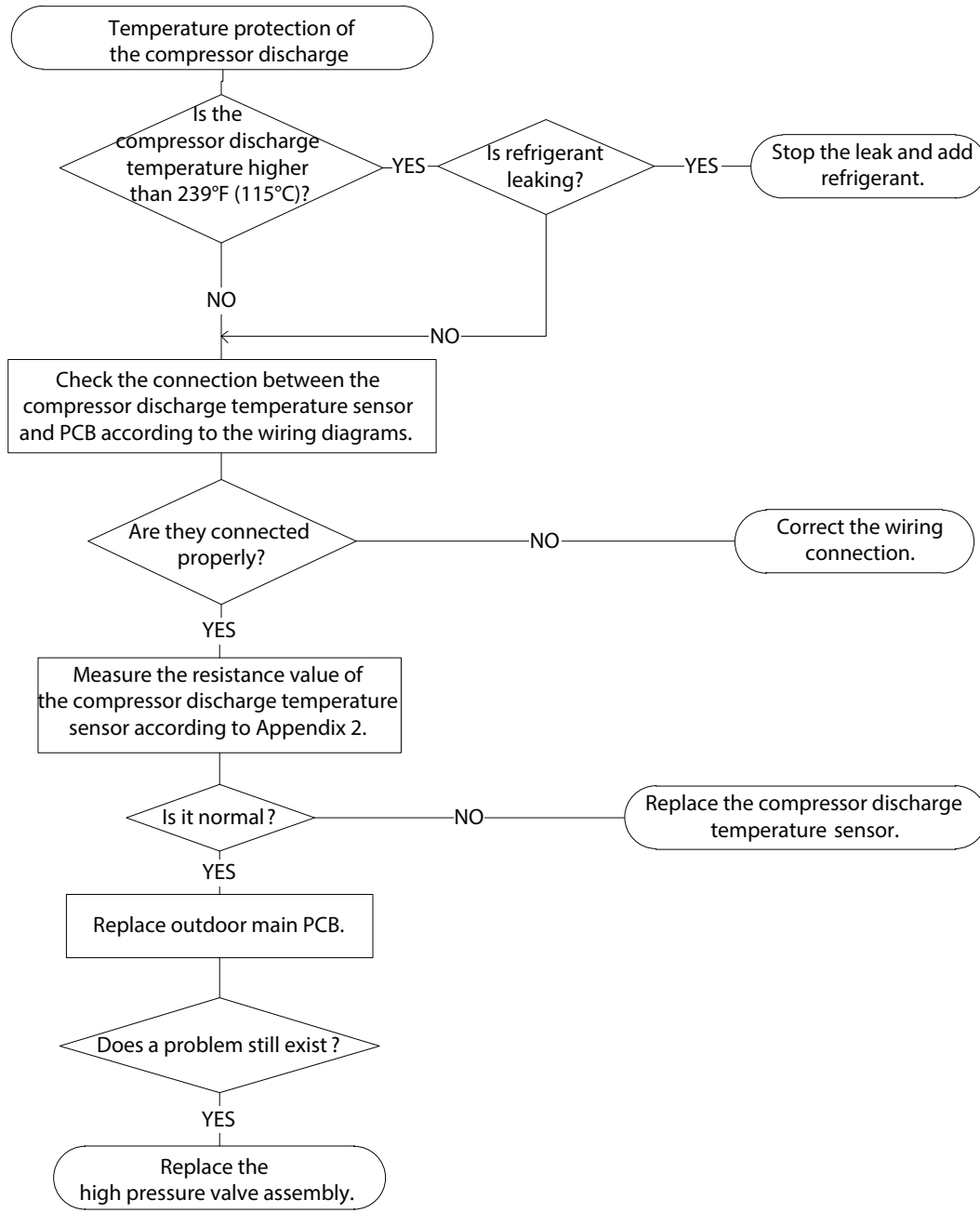
### J2 (High discharge temperature protection)

**Description:** When the compressor discharge temperature (TP) is more than 115°C for 10 seconds, the compressor stops and does not restart until TP is less than 194°F(90°C).

**Recommended parts to prepare:**

- Refrigerant
- Wiring
- Discharge temperature sensor
- Outdoor PCB

### Troubleshooting



## DIAGNOSIS AND SOLUTION (CONT.)

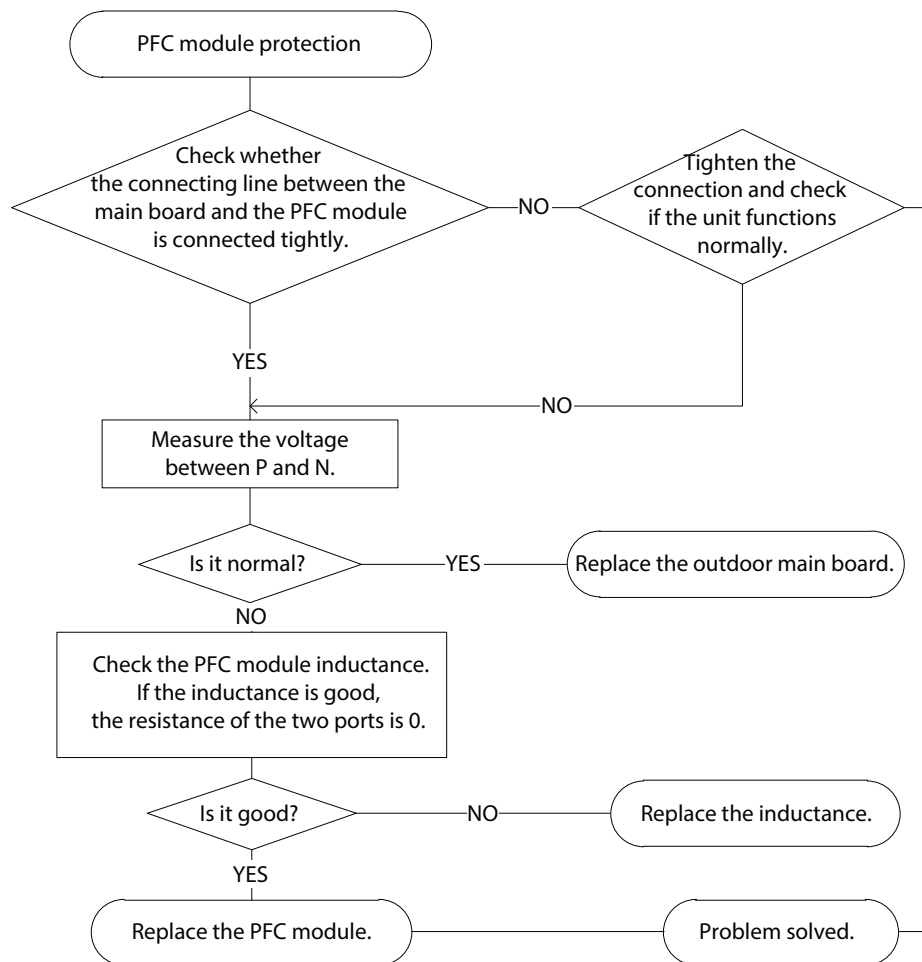
### J3 (PFC module protection)

**Description:** When the voltage signal that the IPM sends to the compressor drive chip is abnormal, the display LED displays “J3” and the air conditioner turns off.

**Recommended parts to prepare:**

- Wiring
- IPM board
- Outdoor fan assembly
- Compressor
- Outdoor PCB

### Troubleshooting



## DIAGNOSIS AND SOLUTION (CONT.)

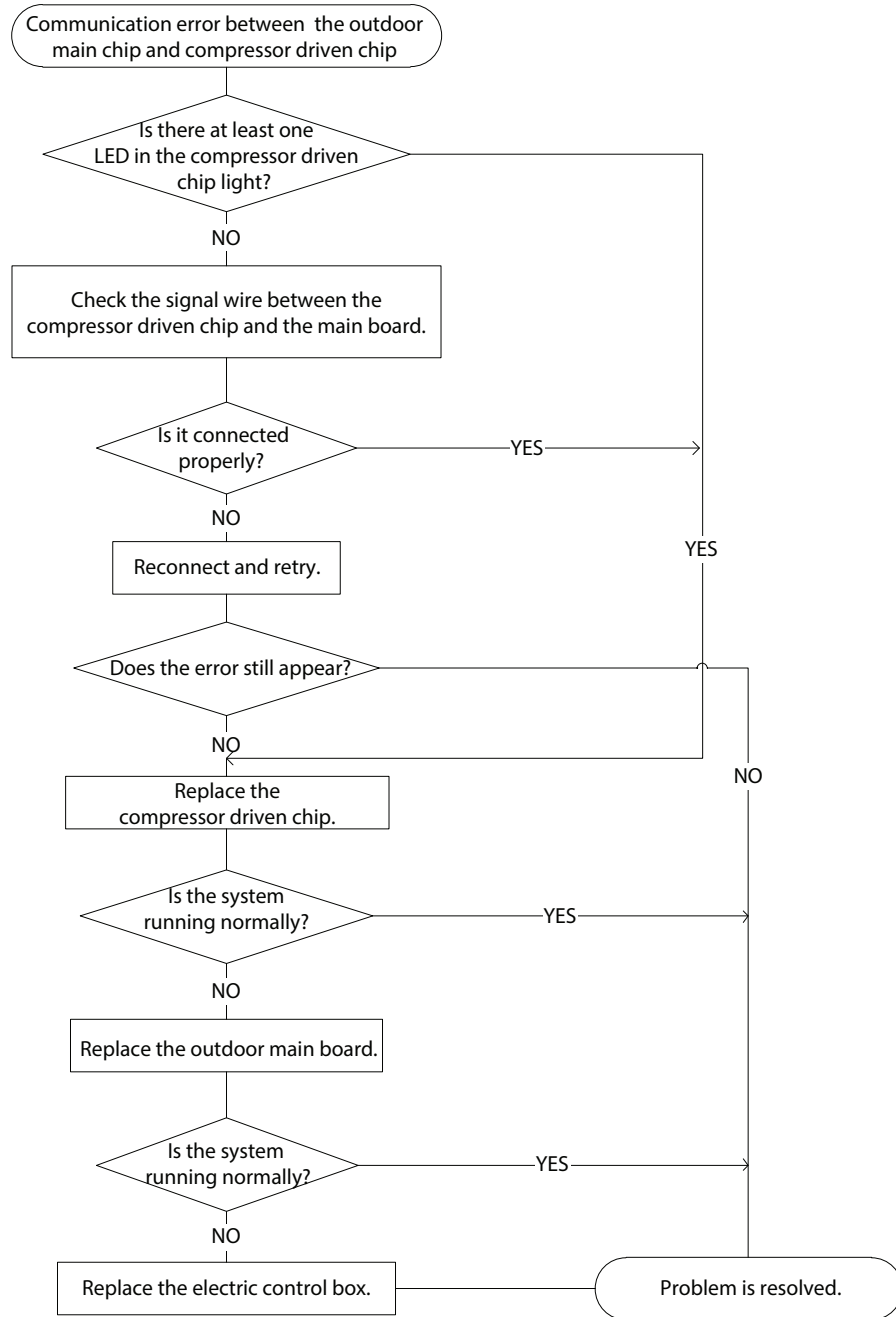
### J4 (Communication error between outdoor main chip and compressor driven chip)

**Description:** The main PCB has not received feedback from the driven chip for one minute or the feedback data is incorrect. The driven chip has not received feedback from the main PCB for one minute or the feedback data is wrong. The failure code disappears after the compressor stops or the communication returns to normal.

**Recommended parts to prepare:**

- Outdoor PCB
- Compressor driven chip
- The signal wire

### Troubleshooting



## DIAGNOSIS AND SOLUTION (CONT)

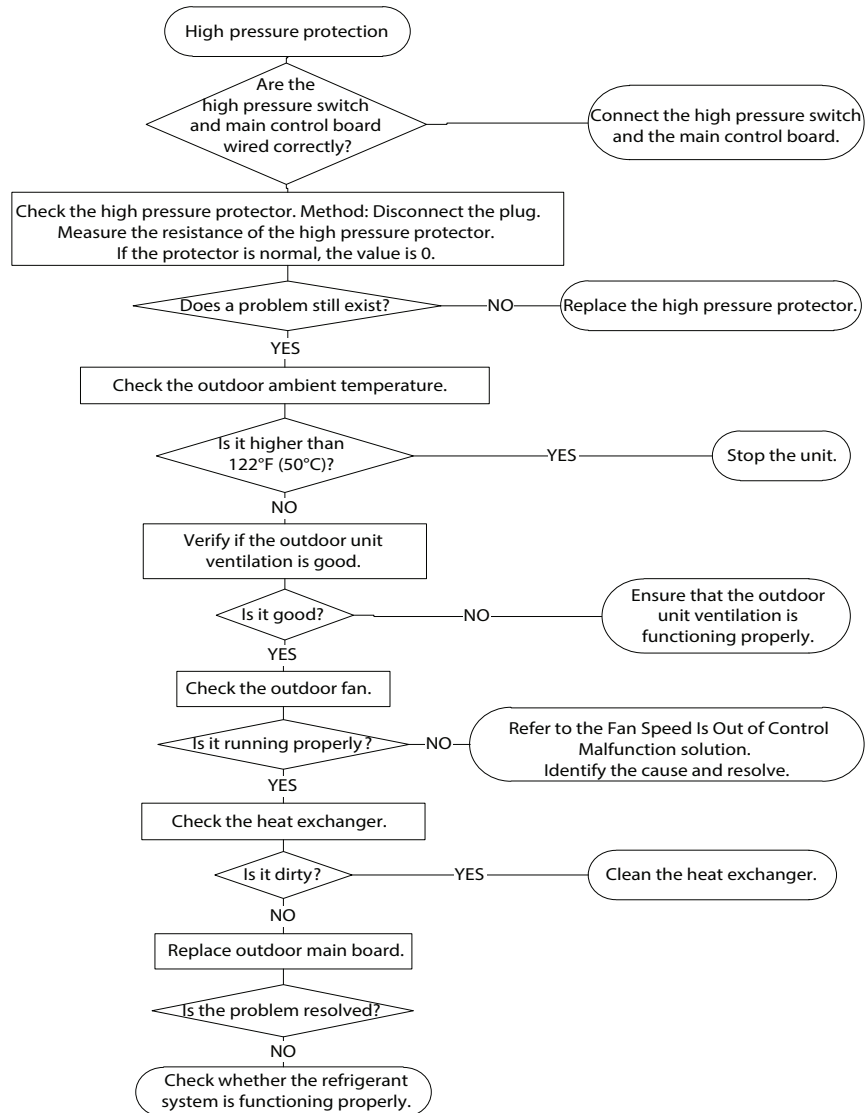
### J5 (High pressure protection)

**Description:** If the sampling voltage is not 5V, the LED displays a failure code.

**Recommended parts to prepare:**

- Wiring
- Overload protector
- Outdoor PCB

### Troubleshooting



## DIAGNOSIS AND SOLUTION (CONT.)

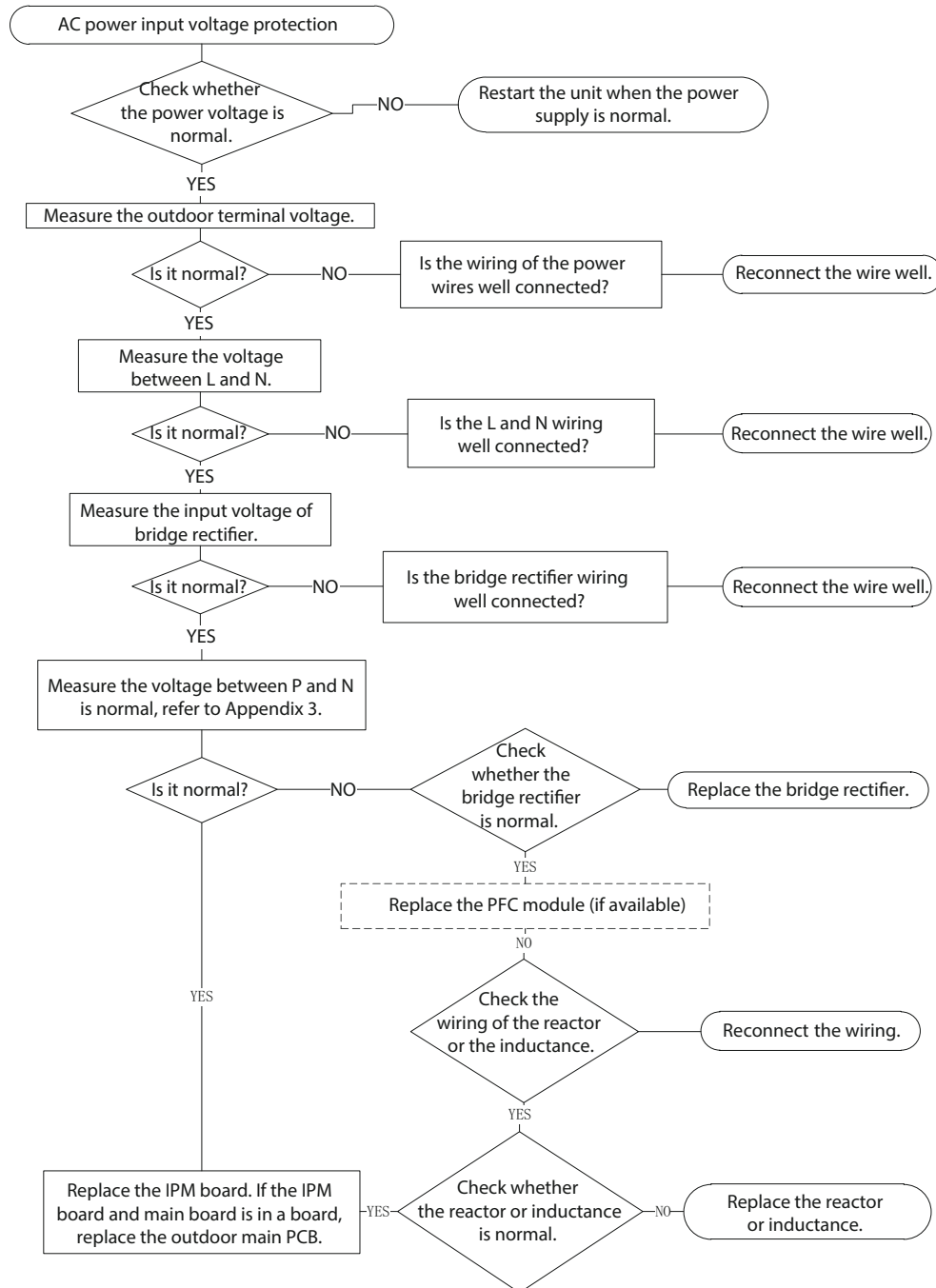
### J8 (AC power input voltage protection)

**Description:** An abnormal voltage rise or drop is detected by checking the specified voltage detection circuit.

**Recommended parts to prepare:**

- Wiring
- Bridge rectifier
- IPM board

### Troubleshooting



## DIAGNOSIS AND SOLUTION (CONT.)

### P5 (Indoor units mode conflict (match with multi outdoor unit))

**Description:** The indoor units cannot work in the COOLING mode and the HEATING mode at the same time because the HEATING mode has the priority.

- If indoor unit A is operating in the COOLING mode or the FAN mode, and indoor unit B is set to the HEATING mode, then unit A powers off and unit B operates in the HEATING mode.
- If indoor unit A is operating in the HEATING mode, and indoor unit B is set to the COOLING mode or FAN mode, then unit B switches to the STANDBY mode and unit A does not change modes.

**Table 11 — Modes**

|              | COOLING MODE | HEATING MODE | FAN | OFF |
|--------------|--------------|--------------|-----|-----|
| COOLING MODE | No           | Yes          | No  | No  |
| HEATING MODE | Yes          | No           | Yes | No  |
| FAN          | No           | Yes          | No  | No  |
| OFF          | No           | No           | No  | No  |

**NOTE:**

- **NO:** No mode conflict
- **YES:** Mode conflict

## DIAGNOSIS AND SOLUTION (CONT.)

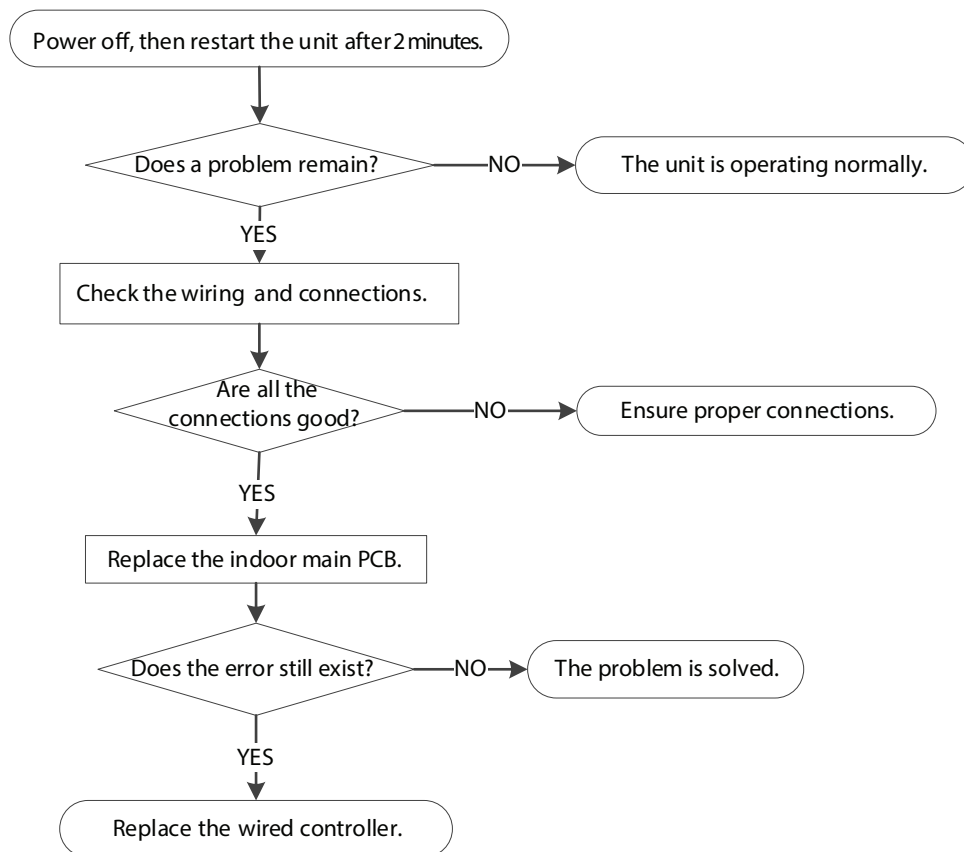
### F0/EH b3 (Communication error between wired controller and indoor unit Diagnosis and Solution)

**Description:** If the indoor PCB does not receive feedback from the wired controller, the error displays on the wired controller.

**Recommended parts to prepare:**

- Connection wires
- Indoor PCB
- Wired controller

#### Troubleshooting



## Check Procedures

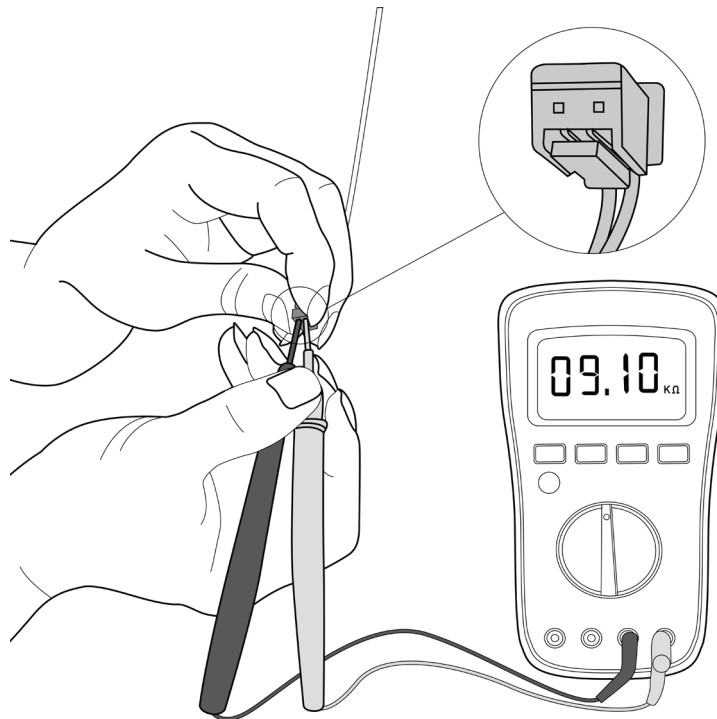
### Temperature Sensor Check



## WARNING

Be sure to turn off all power supplies or disconnect all wires to avoid electric shock. Operate after compressor and coil have returned to normal temperature in case of injury.

1. Disconnect the temperature sensor from the PCB (see “Evaporator” on page 53).
2. Measure the sensor’s resistance value with a multi-meter.
3. Check the corresponding temperature sensor resistance value table (refer to “APPENDICES” on page 55).



Compressor Check

- 1. Disconnect the compressor power cord from outdoor PCB.
- 2. Measure the resistance value of each winding using a multi-meter.
- 3. Check the resistance value of each winding in the Tables 12 - 13.

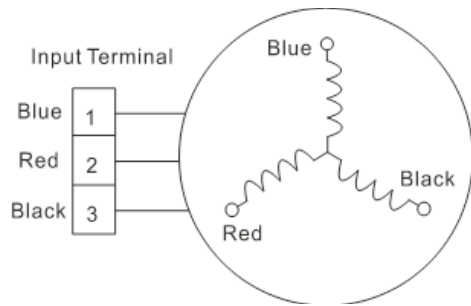
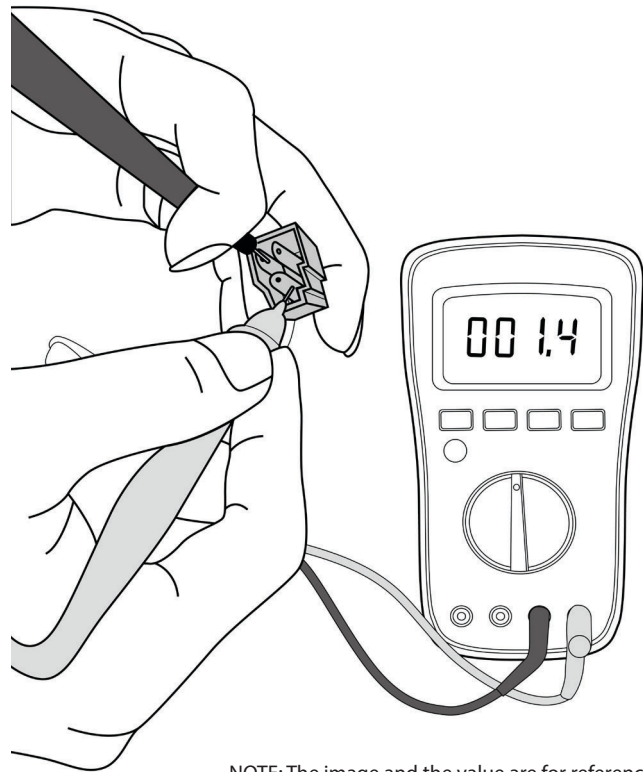


Table 12 — Resistance Value

| RESISTANCE VALUE | KSN98D64UFZ3 | KSN140D21UFZ | KTM240D43UKT | KTM240D57UMT |
|------------------|--------------|--------------|--------------|--------------|
| Blue-Red         | 2.7Ω         | 1.28Ω        | 1.03Ω        | 0.62Ω        |
| Blue-Black       |              |              |              |              |
| Red-Black        |              |              |              |              |

Table 13 — Resistance Value

| RESISTANCE VALUE | KTF250D22UMT | KTF310D43UMT | KTQ420D1UMU<br>EAPQ440D1UMUA | KTN110D42UFZ |
|------------------|--------------|--------------|------------------------------|--------------|
| Blue-Red         | 0.75Ω        | 0.65Ω        | 0.37Ω                        | 1.82Ω        |
| Blue-Black       |              |              |                              |              |
| Red-Black        |              |              |                              |              |



NOTE: The image and the value are for reference only.

IPM Continuity Check



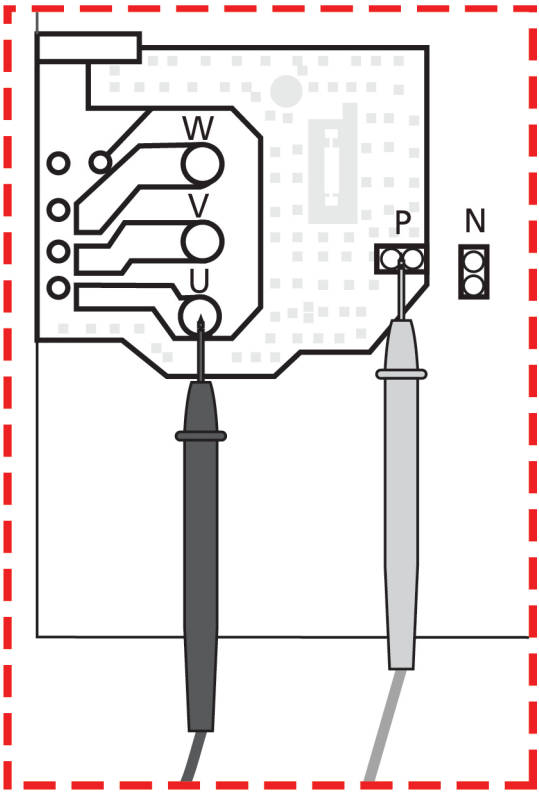
# WARNING

Electricity remains in capacitors even when the power supply is off.  
Ensure the capacitors are fully discharged before troubleshooting.

- 1. Turn off the outdoor unit and disconnect the power supply.
- 2. Discharge the electrolytic capacitors and ensure the energy-storage unit(s) have been discharged.
- 3. Disassemble the outdoor PCB or disassemble the IPM board.
- 4. Measure the resistance value between P and U(V, W, N); U(V, W) and N.

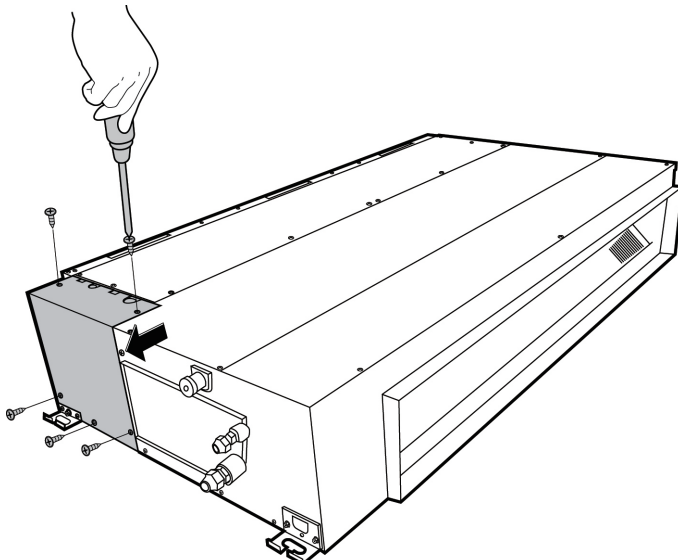
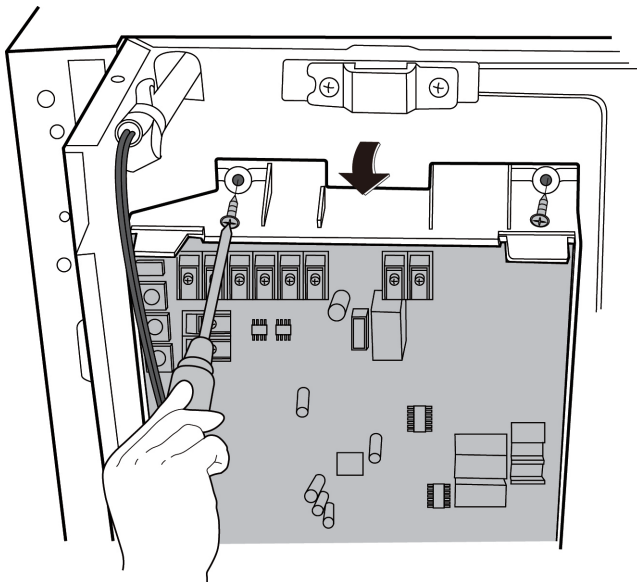
Table 14 — Digital Tester

| DIGITAL TESTER |          | RESISTANCE VALUE       | DIGITAL TESTER |          | RESISTANCE VALUE       |
|----------------|----------|------------------------|----------------|----------|------------------------|
| (+)Red         | (-)Black | $\infty$<br>Several MΩ | (+)Red         | (-)Black | $\infty$<br>Several MΩ |
| P              | N        |                        | U              | N        |                        |
|                | U        |                        | V              |          |                        |
|                | V        |                        | W              |          |                        |
|                | W        |                        | -              |          |                        |



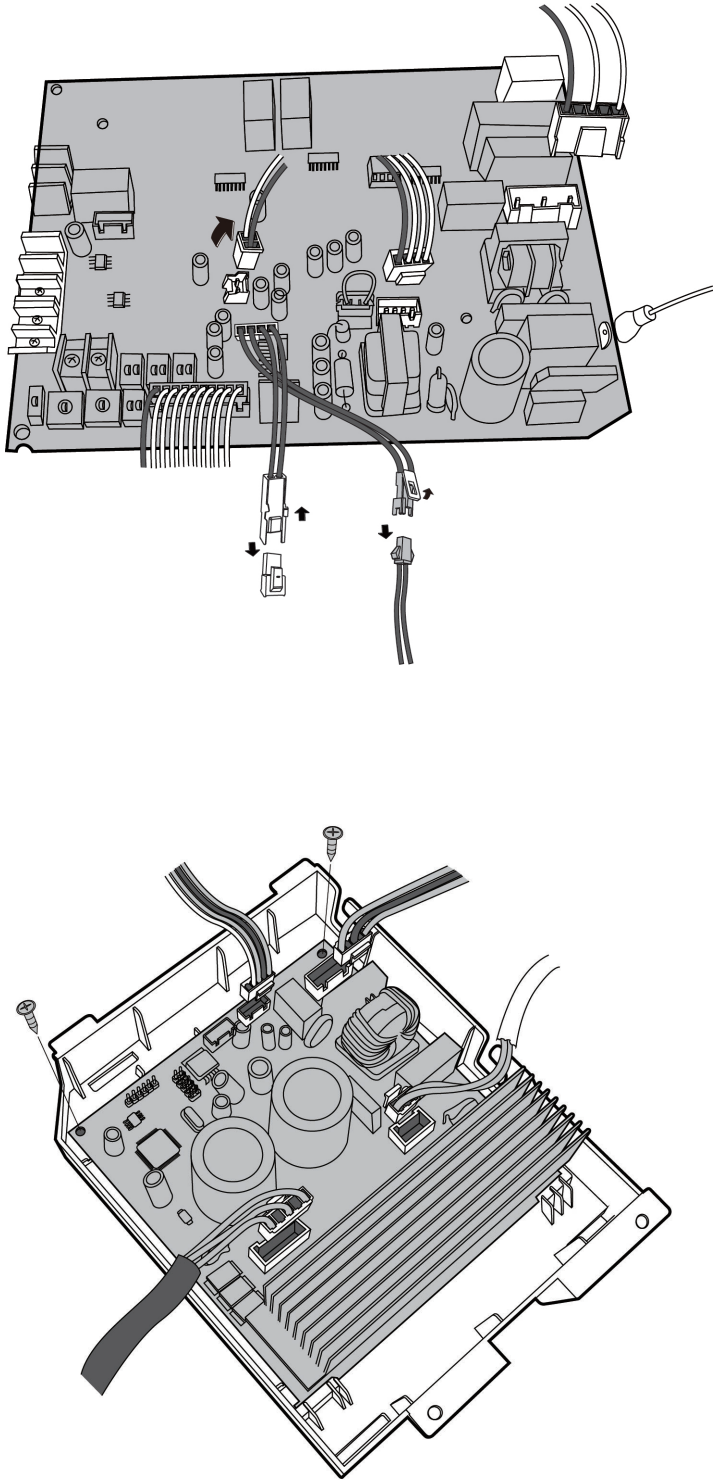
## DISASSEMBLY INSTRUCTIONS

Electrical Parts (Anti static gloves must be worn)

| Procedure                                                                                                                                                                                           | Illustration                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1) Remove the cover screws (5) of the electronic control box then remove the cover.</p> <p>2) Remove 2 screws of the electronic control box. Then release 2 hooks of the main control board.</p> |   |

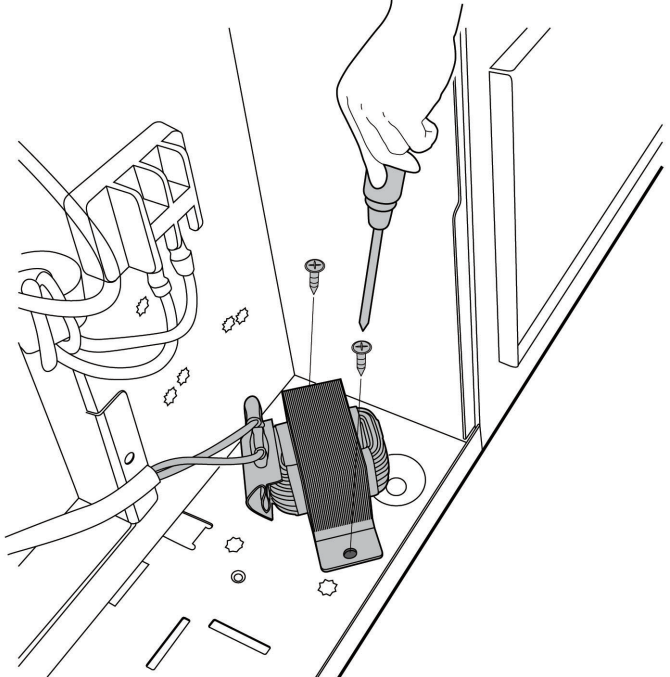
## DISASSEMBLY INSTRUCTIONS (CONT)

### Electrical Parts (Anti static gloves must be worn) (CONT)

| Procedure                                                                                                                                                                                                   | Illustration                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <p>3) Disconnect the connectors then remove the front main control board.</p> <p>4) Turn over the electronic control box. Disconnect the connectors and remove 2 screws of the rear main control board.</p> |  |

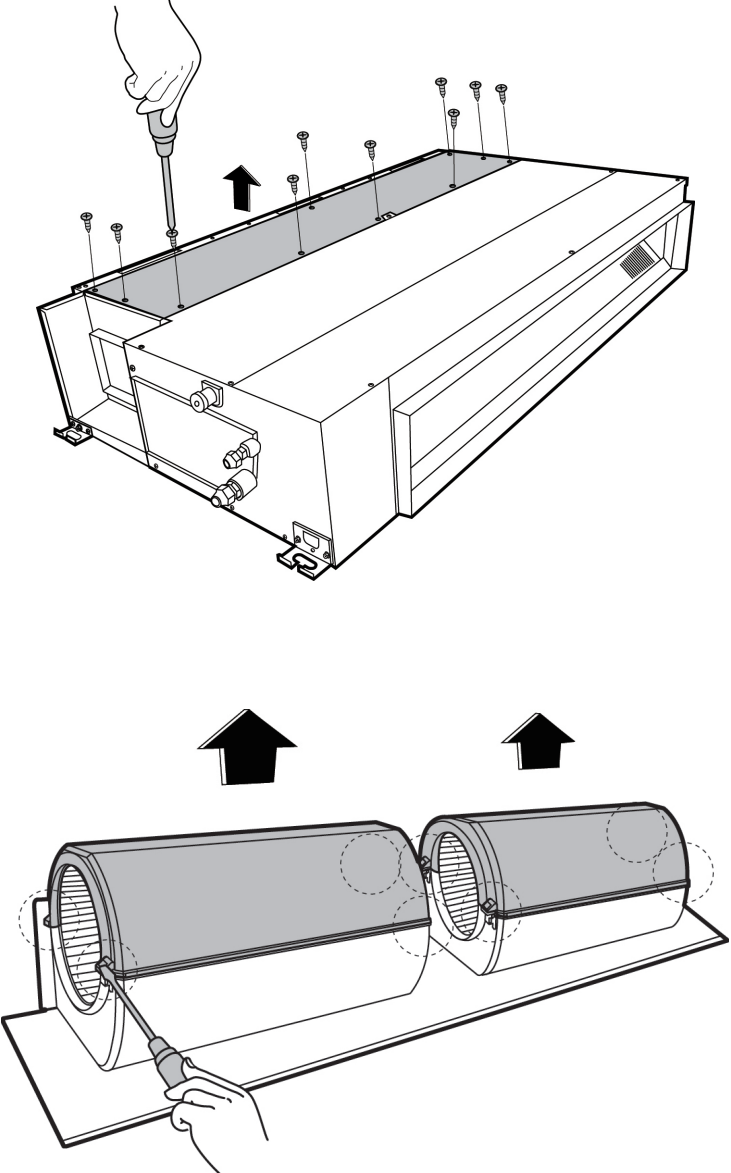
## DISASSEMBLY INSTRUCTIONS (CONT)

### Electrical Parts (Anti static gloves must be worn) (CONT)

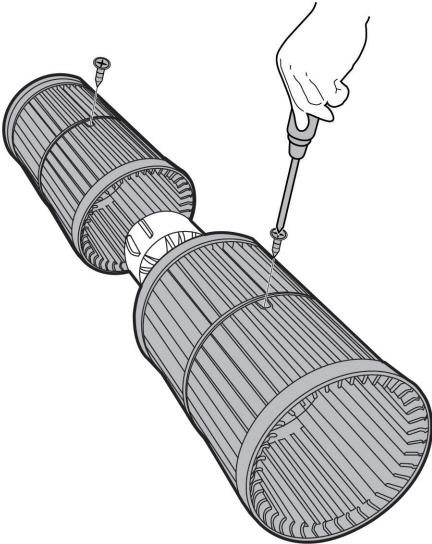
| Procedure                                                        | Illustration                                                                        |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <p>5) Remove 2 screws of the reactor and remove the reactor.</p> |  |

## DISASSEMBLY INSTRUCTIONS (CONT)

### Fan and Motor

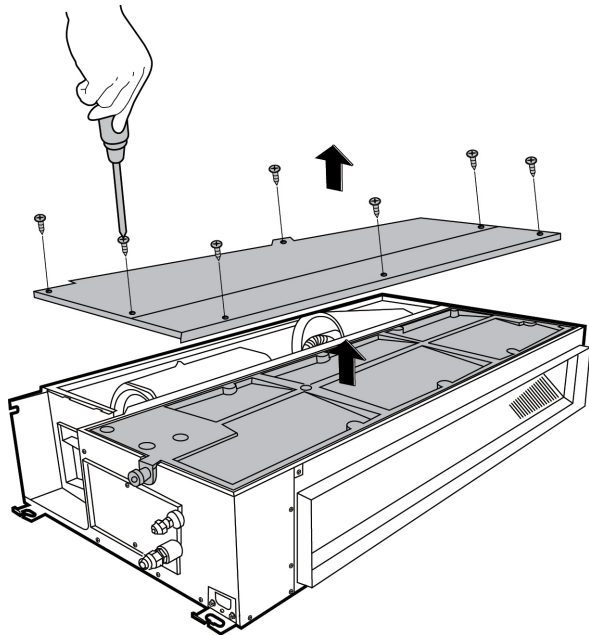
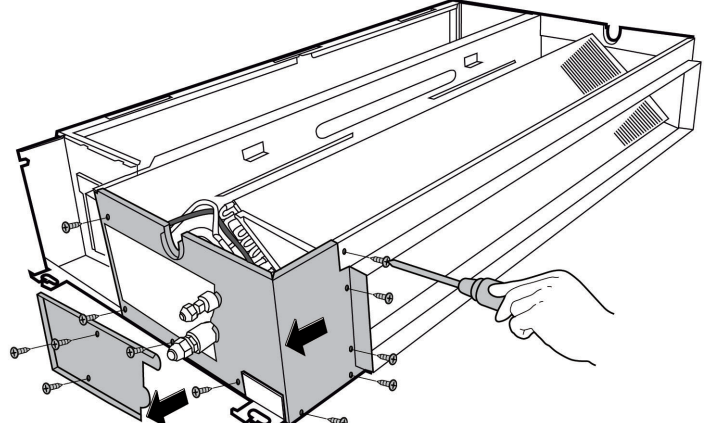
| Procedure                                                       | Illustration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1) Remove 10 top cover screws then remove the top cover.</p> |  <p>The illustration consists of two parts. The top part shows a hand using a screwdriver to remove screws from the top cover of a rectangular unit. An upward arrow indicates the cover is lifted off. The bottom part shows the two halves of the volute shell being separated. A hand is shown using a screwdriver to pry open the shell, with dashed lines indicating the internal components. Two upward arrows indicate the shell halves are lifted away from the unit.</p> |

**DISASSEMBLY INSTRUCTIONS (CONT)****Fan and Motor (CONT)**

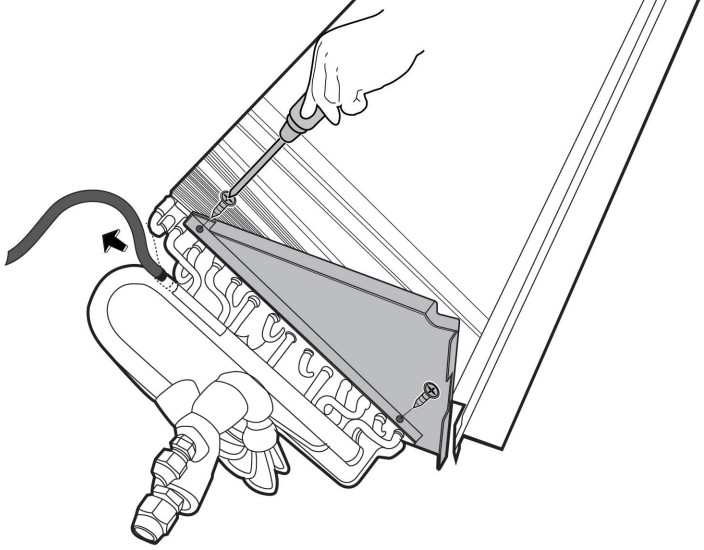
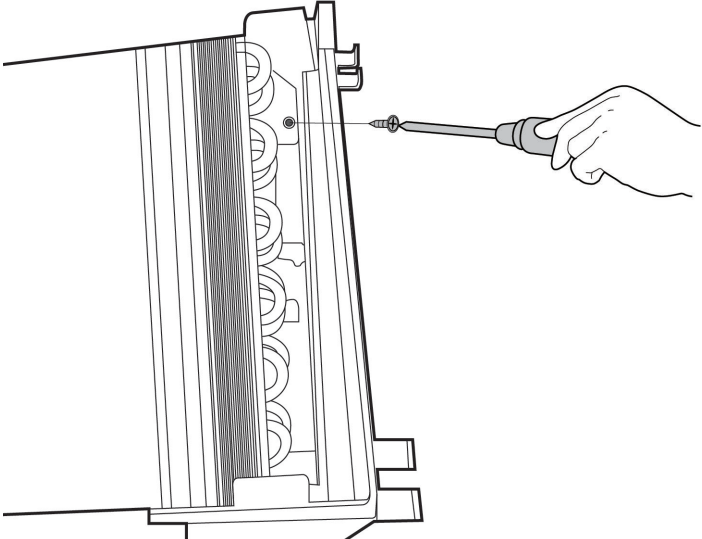
| Procedure                          | Illustration                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>3) Remove the 2 fan screws.</p> |  A line drawing illustration of a fan assembly. The fan is cylindrical with a ribbed outer casing. A hand is shown using a screwdriver to remove a screw from the top of the fan. The screwdriver is positioned vertically, with the tip touching the head of a screw. The fan is tilted at an angle, showing its internal structure and the motor housing. |

## DISASSEMBLY INSTRUCTIONS (CONT)

### Evaporator

| Procedure                                                                                      | Illustration                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1) Remove screws (9) from the water collector then remove the water collector.</p>          |  <p>The illustration shows a top-down view of the evaporator assembly. A hand is using a screwdriver to remove screws (9) from the water collector. An upward arrow indicates the removal direction.</p> |
| <p>2) Remove the 3 screws from the pipe clamp board and 9 screws from the left side board.</p> |  <p>The illustration shows a side view of the evaporator assembly. A hand is using a screwdriver to remove screws from the left side board. Arrows indicate the removal direction.</p>                 |

**DISASSEMBLY INSTRUCTIONS (CONT)****Evaporator (CONT)**

| Procedure                                                                                  | Illustration                                                                         |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <p>3) Remove 2 screws from the evaporator support then pull up the temperature sensor.</p> |    |
| <p>4) Remove the evaporator screw then remove the evaporator.</p>                          |  |

# APPENDICES

## Appendix 1

**Table 15 — Temperature Sensor Resistance Value Table for T1,T2,T3 and T4 (°C – K)**

| °C  | °F | K OHM   | °C | °F  | K OHM   | °C | °F  | K OHM   | °C  | °F  | K OHM   |
|-----|----|---------|----|-----|---------|----|-----|---------|-----|-----|---------|
| -20 | -4 | 115.266 | 20 | 68  | 12.6431 | 60 | 140 | 2.35774 | 100 | 212 | 0.62973 |
| -19 | -2 | 108.146 | 21 | 70  | 12.0561 | 61 | 142 | 2.27249 | 101 | 214 | 0.61148 |
| -18 | 0  | 101.517 | 22 | 72  | 11.5    | 62 | 144 | 2.19073 | 102 | 216 | 0.59386 |
| -17 | 1  | 96.3423 | 23 | 73  | 10.9731 | 63 | 145 | 2.11241 | 103 | 217 | 0.57683 |
| -16 | 3  | 89.5865 | 24 | 75  | 10.4736 | 64 | 147 | 2.03732 | 104 | 219 | 0.56038 |
| -15 | 5  | 84.219  | 25 | 77  | 10      | 65 | 149 | 1.96532 | 105 | 221 | 0.54448 |
| -14 | 7  | 79.311  | 26 | 79  | 9.55074 | 66 | 151 | 1.89627 | 106 | 223 | 0.52912 |
| -13 | 9  | 74.536  | 27 | 81  | 9.12445 | 67 | 153 | 1.83003 | 107 | 225 | 0.51426 |
| -12 | 10 | 70.1698 | 28 | 82  | 8.71983 | 68 | 154 | 1.76647 | 108 | 226 | 0.49989 |
| -11 | 12 | 66.0898 | 29 | 84  | 8.33566 | 69 | 156 | 1.70547 | 109 | 228 | 0.486   |
| -10 | 14 | 62.2756 | 30 | 86  | 7.97078 | 70 | 158 | 1.64691 | 110 | 230 | 0.47256 |
| -9  | 16 | 58.7079 | 31 | 88  | 7.62411 | 71 | 160 | 1.59068 | 111 | 232 | 0.45957 |
| -8  | 18 | 56.3694 | 32 | 90  | 7.29464 | 72 | 162 | 1.53668 | 112 | 234 | 0.44699 |
| -7  | 19 | 52.2438 | 33 | 91  | 6.98142 | 73 | 163 | 1.48481 | 113 | 235 | 0.43482 |
| -6  | 21 | 49.3161 | 34 | 93  | 6.68355 | 74 | 165 | 1.43498 | 114 | 237 | 0.42304 |
| -5  | 23 | 46.5725 | 35 | 95  | 6.40021 | 75 | 167 | 1.38703 | 115 | 239 | 0.41164 |
| -4  | 25 | 44      | 36 | 97  | 6.13059 | 76 | 169 | 1.34105 | 116 | 241 | 0.4006  |
| -3  | 27 | 41.5878 | 37 | 99  | 5.87359 | 77 | 171 | 1.29078 | 117 | 243 | 0.38991 |
| -2  | 28 | 39.8239 | 38 | 100 | 5.62961 | 78 | 172 | 1.25423 | 118 | 244 | 0.37956 |
| -1  | 30 | 37.1988 | 39 | 102 | 5.39689 | 79 | 174 | 1.2133  | 119 | 246 | 0.36954 |
| 0   | 32 | 35.2024 | 40 | 104 | 5.17519 | 80 | 176 | 1.17393 | 120 | 248 | 0.35982 |
| 1   | 34 | 33.3269 | 41 | 106 | 4.96392 | 81 | 178 | 1.13604 | 121 | 250 | 0.35042 |
| 2   | 36 | 31.5635 | 42 | 108 | 4.76253 | 82 | 180 | 1.09958 | 122 | 252 | 0.3413  |
| 3   | 37 | 29.9058 | 43 | 109 | 4.5705  | 83 | 181 | 1.06448 | 123 | 253 | 0.33246 |
| 4   | 39 | 28.3459 | 44 | 111 | 4.38736 | 84 | 183 | 1.03069 | 124 | 255 | 0.3239  |
| 5   | 41 | 26.8778 | 45 | 113 | 4.21263 | 85 | 185 | 0.99815 | 125 | 257 | 0.31559 |
| 6   | 43 | 25.4954 | 46 | 115 | 4.04589 | 86 | 187 | 0.96681 | 126 | 259 | 0.30754 |
| 7   | 45 | 24.1932 | 47 | 117 | 3.88673 | 87 | 189 | 0.93662 | 127 | 261 | 0.29974 |
| 8   | 46 | 22.5662 | 48 | 118 | 3.73476 | 88 | 190 | 0.90753 | 128 | 262 | 0.29216 |
| 9   | 48 | 21.8094 | 49 | 120 | 3.58962 | 89 | 192 | 0.8795  | 129 | 264 | 0.28482 |
| 10  | 50 | 20.7184 | 50 | 122 | 3.45097 | 90 | 194 | 0.85248 | 130 | 266 | 0.2777  |
| 11  | 52 | 19.6891 | 51 | 124 | 3.31847 | 91 | 196 | 0.82643 | 131 | 268 | 0.27078 |
| 12  | 54 | 18.7177 | 52 | 126 | 3.19183 | 92 | 198 | 0.80132 | 132 | 270 | 0.26408 |
| 13  | 55 | 17.8005 | 53 | 127 | 3.07075 | 93 | 199 | 0.77709 | 133 | 271 | 0.25757 |
| 14  | 57 | 16.9341 | 54 | 129 | 2.95896 | 94 | 201 | 0.75373 | 134 | 273 | 0.25125 |
| 15  | 59 | 16.1156 | 55 | 131 | 2.84421 | 95 | 203 | 0.73119 | 135 | 275 | 0.24512 |
| 16  | 61 | 15.3418 | 56 | 133 | 2.73823 | 96 | 205 | 0.70944 | 136 | 277 | 0.23916 |
| 17  | 63 | 14.6181 | 57 | 135 | 2.63682 | 97 | 207 | 0.68844 | 137 | 279 | 0.23338 |
| 18  | 64 | 13.918  | 58 | 136 | 2.53973 | 98 | 208 | 0.66818 | 138 | 280 | 0.22776 |
| 19  | 66 | 13.2631 | 59 | 138 | 2.44677 | 99 | 210 | 0.64862 | 139 | 282 | 0.22231 |

## Appendix 2

**Table 16 — Temperature Sensor Resistance Value Table for TP (for some units) (°C--K)**

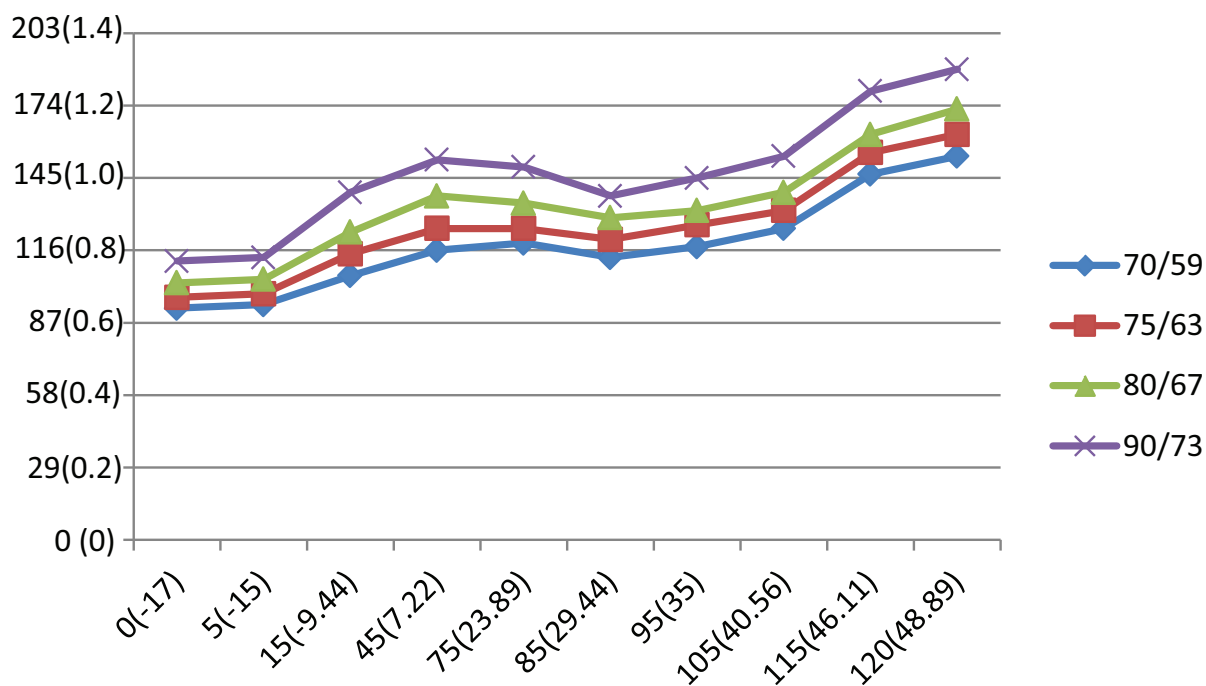
| °C  | °F | K OHM | °C | °F  | K OHM | °C | °F  | K OHM | °C  | °F  | K OHM |
|-----|----|-------|----|-----|-------|----|-----|-------|-----|-----|-------|
| -20 | -4 | 542.7 | 20 | 68  | 68.66 | 60 | 140 | 13.59 | 100 | 212 | 3.702 |
| -19 | -2 | 511.9 | 21 | 70  | 65.62 | 61 | 142 | 13.11 | 101 | 214 | 3.595 |
| -18 | 0  | 483   | 22 | 72  | 62.73 | 62 | 144 | 12.65 | 102 | 216 | 3.492 |
| -17 | 1  | 455.9 | 23 | 73  | 59.98 | 63 | 145 | 12.21 | 103 | 217 | 3.392 |
| -16 | 3  | 430.5 | 24 | 75  | 57.37 | 64 | 147 | 11.79 | 104 | 219 | 3.296 |
| -15 | 5  | 406.7 | 25 | 77  | 54.89 | 65 | 149 | 11.38 | 105 | 221 | 3.203 |
| -14 | 7  | 384.3 | 26 | 79  | 52.53 | 66 | 151 | 10.99 | 106 | 223 | 3.113 |
| -13 | 9  | 363.3 | 27 | 81  | 50.28 | 67 | 153 | 10.61 | 107 | 225 | 3.025 |
| -12 | 10 | 343.6 | 28 | 82  | 48.14 | 68 | 154 | 10.25 | 108 | 226 | 2.941 |
| -11 | 12 | 325.1 | 29 | 84  | 46.11 | 69 | 156 | 9.902 | 109 | 228 | 2.86  |
| -10 | 14 | 307.7 | 30 | 86  | 44.17 | 70 | 158 | 9.569 | 110 | 230 | 2.781 |
| -9  | 16 | 291.3 | 31 | 88  | 42.33 | 71 | 160 | 9.248 | 111 | 232 | 2.704 |
| -8  | 18 | 275.9 | 32 | 90  | 40.57 | 72 | 162 | 8.94  | 112 | 234 | 2.63  |
| -7  | 19 | 261.4 | 33 | 91  | 38.89 | 73 | 163 | 8.643 | 113 | 235 | 2.559 |
| -6  | 21 | 247.8 | 34 | 93  | 37.3  | 74 | 165 | 8.358 | 114 | 237 | 2.489 |
| -5  | 23 | 234.9 | 35 | 95  | 35.78 | 75 | 167 | 8.084 | 115 | 239 | 2.422 |
| -4  | 25 | 222.8 | 36 | 97  | 34.32 | 76 | 169 | 7.82  | 116 | 241 | 2.357 |
| -3  | 27 | 211.4 | 37 | 99  | 32.94 | 77 | 171 | 7.566 | 117 | 243 | 2.294 |
| -2  | 28 | 200.7 | 38 | 100 | 31.62 | 78 | 172 | 7.321 | 118 | 244 | 2.233 |
| -1  | 30 | 190.5 | 39 | 102 | 30.36 | 79 | 174 | 7.086 | 119 | 246 | 2.174 |
| 0   | 32 | 180.9 | 40 | 104 | 29.15 | 80 | 176 | 6.859 | 120 | 248 | 2.117 |
| 1   | 34 | 171.9 | 41 | 106 | 28    | 81 | 178 | 6.641 | 121 | 250 | 2.061 |
| 2   | 36 | 163.3 | 42 | 108 | 26.9  | 82 | 180 | 6.43  | 122 | 252 | 2.007 |
| 3   | 37 | 155.2 | 43 | 109 | 25.86 | 83 | 181 | 6.228 | 123 | 253 | 1.955 |
| 4   | 39 | 147.6 | 44 | 111 | 24.85 | 84 | 183 | 6.033 | 124 | 255 | 1.905 |
| 5   | 41 | 140.4 | 45 | 113 | 23.89 | 85 | 185 | 5.844 | 125 | 257 | 1.856 |
| 6   | 43 | 133.5 | 46 | 115 | 22.89 | 86 | 187 | 5.663 | 126 | 259 | 1.808 |
| 7   | 45 | 127.1 | 47 | 117 | 22.1  | 87 | 189 | 5.488 | 127 | 261 | 1.762 |
| 8   | 46 | 121   | 48 | 118 | 21.26 | 88 | 190 | 5.32  | 128 | 262 | 1.717 |
| 9   | 48 | 115.2 | 49 | 120 | 20.46 | 89 | 192 | 5.157 | 129 | 264 | 1.674 |
| 10  | 50 | 109.8 | 50 | 122 | 19.69 | 90 | 194 | 5     | 130 | 266 | 1.632 |
| 11  | 52 | 104.6 | 51 | 124 | 18.96 | 91 | 196 | 4.849 |     |     |       |
| 12  | 54 | 99.69 | 52 | 126 | 18.26 | 92 | 198 | 4.703 |     |     |       |
| 13  | 55 | 95.05 | 53 | 127 | 17.58 | 93 | 199 | 4.562 |     |     |       |
| 14  | 57 | 90.66 | 54 | 129 | 16.94 | 94 | 201 | 4.426 |     |     |       |
| 15  | 59 | 86.49 | 55 | 131 | 16.32 | 95 | 203 | 4.294 |     |     |       |
| 16  | 61 | 82.54 | 56 | 133 | 15.73 | 96 | 205 | 4.167 |     |     |       |
| 17  | 63 | 78.79 | 57 | 135 | 15.16 | 97 | 207 | 4.045 |     |     |       |
| 18  | 64 | 75.24 | 58 | 136 | 14.62 | 98 | 208 | 3.927 |     |     |       |
| 19  | 66 | 71.86 | 59 | 138 | 14.09 | 99 | 210 | 3.812 |     |     |       |

## Pressure on the Service Port

### Cooling Chart (R410A)

Table 17 — Cooling Chart

| °F(°C) | ODU(DB)             | 0     | 5     | 15      | 45     | 75      | 85      | 95   | 105     | 115     | 120     |
|--------|---------------------|-------|-------|---------|--------|---------|---------|------|---------|---------|---------|
|        | IDU(DB/WB)          | (-17) | (-15) | (-9.44) | (7.22) | (23.89) | (29.44) | (35) | (40.56) | (46.11) | (48.89) |
| PSI    | 70/59 (21.11/15)    | 93    | 94    | 106     | 116    | 119     | 113     | 117  | 125     | 147     | 154     |
|        | 75/63 (23.89/17.22) | 97    | 99    | 115     | 125    | 124     | 120     | 126  | 132     | 155     | 162     |
|        | 80/67 (26.67/19.44) | 103   | 104   | 123     | 138    | 135     | 129     | 132  | 140     | 162     | 173     |
|        | 90/73 (32.22/22.78) | 112   | 113   | 139     | 152    | 149     | 138     | 145  | 154     | 180     | 189     |
| MPa    | 70/59 (21.11/15)    | 0.64  | 0.65  | 0.73    | 0.8    | 0.82    | 0.78    | 0.81 | 0.86    | 1.01    | 1.06    |
|        | 75/63 (23.89/17.22) | 0.67  | 0.68  | 0.79    | 0.86   | 0.86    | 0.83    | 0.87 | 0.91    | 1.07    | 1.12    |
|        | 80/67 (26.67/19.44) | 0.71  | 0.72  | 0.85    | 0.95   | 0.93    | 0.89    | 0.91 | 0.96    | 1.12    | 1.19    |
|        | 90/73 (32.22/22.78) | 0.77  | 0.78  | 0.96    | 1.05   | 1.03    | 0.95    | 1    | 1.06    | 1.24    | 1.3     |



**Heating Chart (R410A)****Table 18 — Heating Chart**

| °F(°C) | ODU(DB/WB) | 57/53<br>(13.89/11.67) | 47/43<br>(8.33/6.11) | 37/33<br>(2.78/0.56) | 27/23<br>(-2.78/-5) | 17/13<br>(-8.33/-10.56) | 0/-2<br>(-17/-19) | -17/-18<br>(-27/-28) |
|--------|------------|------------------------|----------------------|----------------------|---------------------|-------------------------|-------------------|----------------------|
|        | IDU(DB)    |                        |                      |                      |                     |                         |                   |                      |
| PSI    | 55(12.78)  | 439                    | 413                  | 367                  | 330                 | 302                     | 268               | 239                  |
|        | 65(18.33)  | 471                    | 435                  | 386                  | 368                 | 339                     | 297               | 276                  |
|        | 75(23.89)  | 489                    | 457                  | 403                  | 381                 | 362                     | 312               | 290                  |
| MPa    | 55(12.78)  | 3.03                   | 2.85                 | 2.53                 | 2.28                | 2.08                    | 1.85              | 1.65                 |
|        | 65(18.33)  | 3.25                   | 3.00                 | 2.66                 | 2.54                | 2.33                    | 2.05              | 1.90                 |
|        | 75(23.89)  | 3.38                   | 3.15                 | 2.78                 | 2.63                | 2.49                    | 2.15              | 2.00                 |

